

Windows 10 Client Deployment & Systems Administration

Virtual Machine Installation, System Utilities, Storage Configuration, User & Group Management, and Security Policy Hardening

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COMPETENCY LEVEL

Foundation — Client Systems Administration

Hyper-V Virtualization	System Utilities	Device Manager
Storage Spaces	Disk Management	Local Users & Groups
Local Group Policy	Security Hardening	

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01

PROJECT OVERVIEW

Objective

This portfolio documents the **deployment, configuration, and administration** of two Windows 10 Enterprise Evaluation client operating systems — `Tiisetso_Win10A` and `Tiisetso_Win10B` — built and validated inside a virtualised Hyper-V lab environment. The brief required a single, coherent body of evidence spanning five practical disciplines: virtual machine provisioning and OS installation, native system utilities and troubleshooting tools, local storage and disk management, local user and group administration, and security policy hardening through Local Group Policy.

The objective throughout was to demonstrate not just that each native Windows 10 tool could be operated, but **why** a given configuration choice was made — treating each of the two client machines as representative endpoints that a systems administrator would be responsible for deploying, securing, and troubleshooting in a live environment.

Scope & Context

The work was carried out inside Hyper-V Manager on a lab host, with two Generation 2 virtual machines provisioned from a shared Windows 10 Enterprise Evaluation ISO. Each VM was installed, localised to South African regional settings, and renamed to identify its role (`Tiisetso_Win10A`, `Tiisetso_Win10B`) before device-specific configuration diverged: **Win10A** was used to demonstrate two-way mirror Storage Spaces resiliency and core installation evidence, while **Win10B** was used to demonstrate parity Storage Spaces resiliency, local user and group administration, and the full Local Group Policy security baseline.

Deliverables Covered in this Portfolio

VM Deployment & Installation	Two Generation 2 Hyper-V virtual machines provisioned, installed from a shared Windows 10 Enterprise Evaluation ISO, localised to South African settings, and renamed to identify each machine's role.
System Utilities & Troubleshooting	Task Scheduler automation for disk maintenance, Driver Verifier Manager signature auditing, and Device Manager-based hardware troubleshooting.
Storage & Disk Management	Storage Spaces pools configured with both parity and two-way mirror resiliency, dynamic volume verification, and Hyper-V virtual hard disk provisioning.
Users & Group Management	Local user accounts created and organised into department-aligned local security groups, with privilege escalation applied through group-based Administrator membership.
Security Policy Configuration	A Local Group Policy baseline covering account lockout, password complexity, User Account Control hardening, interactive logon protections, and a legal logon banner — verified against a live Windows Update check and a policy-applied restart.

Outcome

All planned configuration was implemented and verified through direct evidence captured on each virtual machine: a working multi-VM Hyper-V lab; native troubleshooting and maintenance tooling correctly configured; two independently demonstrated Storage Spaces resiliency models; a local identity structure of

users and groups with department-aligned privilege; and an enforced local security baseline confirmed by a live system restart. The following sections present this work as a structured evidence log, followed by a summary of the competencies demonstrated.

Before provisioning began, the platform itself was reviewed against its editions, hardware requirements, licensing model, and feature set, so that the edition and configuration chosen for the lab environment reflected a deliberate decision rather than a default.

Editions

Microsoft offers several editions of Windows 10 to serve different user segments. **Windows 10 Home** targets personal use on desktops and laptops with basic functionality. **Windows 10 Pro** is aimed at small businesses and power users, adding advanced security, remote access, and device management features. **Windows 10 Enterprise** is built for large organisations, layering in additional security and management capabilities on top of Pro. **Windows 10 Education** is a variant of Enterprise licensed for academic institutions. The lab environment in this portfolio uses the **Enterprise Evaluation** build, which exposes the same management and security surface (Group Policy, BitLocker, Storage Spaces) that a production enterprise deployment would use, without requiring a full volume license for a time-boxed lab.

Hardware Requirements

Microsoft's minimum specification for Windows 10 was used as the baseline when sizing the lab virtual machines:

- › Processor: 1 GHz or faster CPU
- › RAM: 1 GB for 32-bit, or 2 GB for 64-bit
- › Storage: 16 GB for 32-bit, or 20 GB for 64-bit
- › Graphics: DirectX 9 or later with a WDDM 1.0 driver
- › Display: 800 × 600 resolution or higher

Both lab VMs were provisioned well above this floor (4096 MB RAM, Generation 2 with a dynamically expanding VHD), since the minimum specification is a functional floor rather than a comfortable working configuration — Storage Spaces and Group Policy tooling in particular benefit from headroom beyond the bare minimum.

Licensing, Activation & Upgrade Pathways

Windows 10 requires activation using either a valid product key or a digital license tied to a Microsoft account, with licensing options spanning retail, volume licensing, and subscription channels such as Microsoft 365. For organisations running older Windows versions, Microsoft provides an upgrade path via Windows Update or a standalone upgrade tool — with data backup and hardware-compatibility checks recommended before initiating the process, since an in-place upgrade carries more risk than a clean install.

Platform Capabilities

A number of platform features are directly relevant to the administrative work carried out later in this portfolio:

- › **Windows Hello** — biometric authentication via facial recognition or fingerprint, relevant to the identity-hardening work in Section 07.
- › **Storage Spaces** — software-defined storage resiliency, demonstrated in Section 05.
- › **Virtual Desktops** — multiple managed desktop workspaces for better task organisation.
- › **Universal Apps** — a consistent application model across device form factors.

Selection Rationale

Windows 10 Enterprise was selected for this lab as the edition best matched to demonstrating administrator-level capability: it is the only edition that exposes the full Local Group Policy, Storage Spaces, and device-management surface used throughout Sections 03–07, while remaining compatible with the same hardware footprint as Home or Pro. This makes it the natural choice both for the technical work in this portfolio and for the kind of managed desktop environment a systems administrator would encounter in production.

03

VIRTUAL MACHINE DEPLOYMENT & INSTALLATION

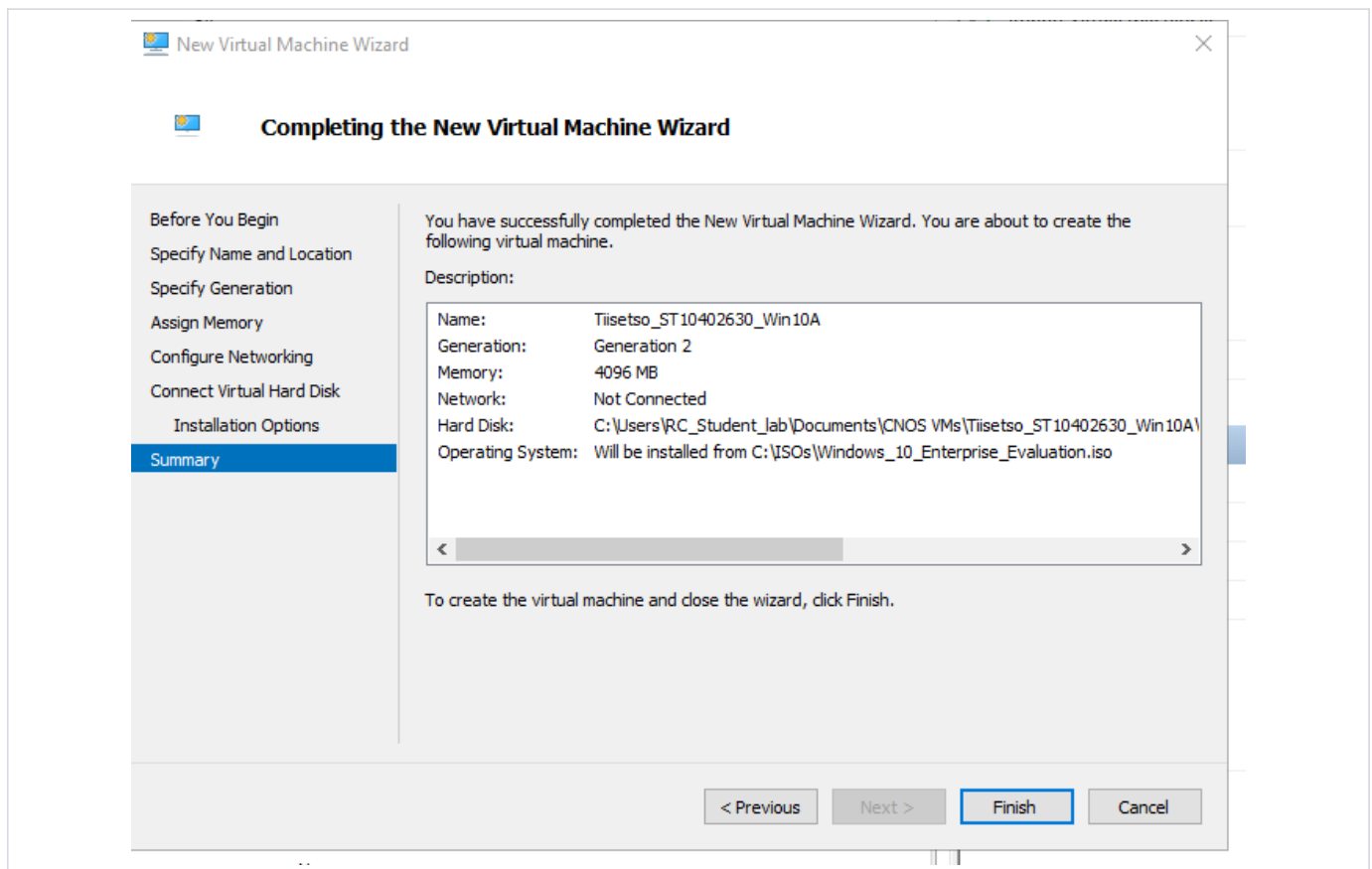
This section covers the provisioning of both client virtual machines in Hyper-V Manager, through to a completed, correctly named Windows 10 installation on each — the foundation that every later section builds on.

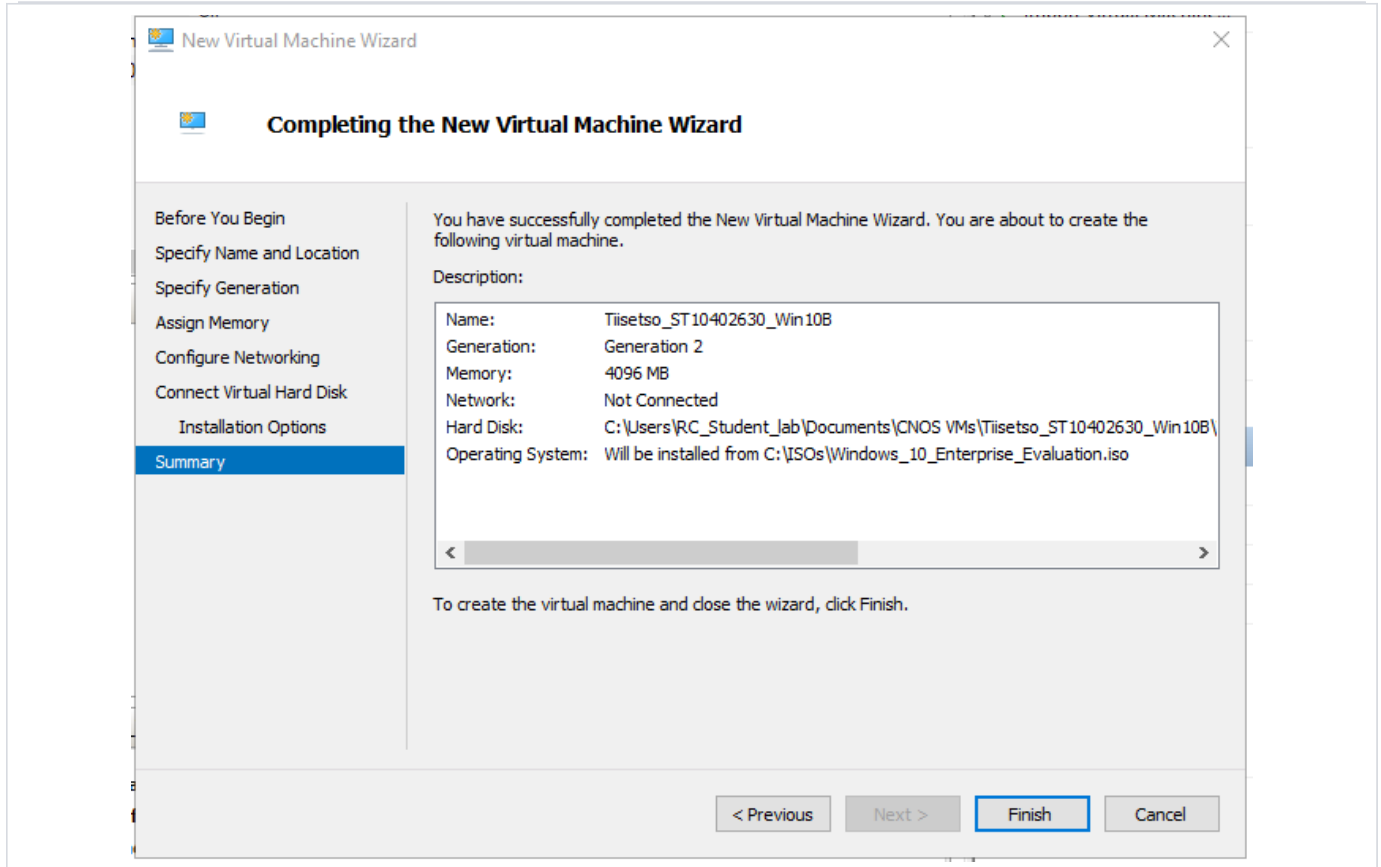
3.1 — Virtual Machine Provisioning

Two Generation 2 virtual machines were created through the New Virtual Machine Wizard — `Tiisetso_ST10402630_Win10A` and `Tiisetso_ST10402630_Win10B` — each assigned 4096 MB of memory and a dynamically expanding virtual hard disk, with the OS source pointed at a shared `Windows_10_Enterprise_Evaluation.iso`. **Generation 2** was chosen over Generation 1 because it supports UEFI boot, Secure Boot, and larger boot volumes — the platform baseline a production Windows 10 Enterprise deployment would run on, rather than the legacy BIOS-emulation path Generation 1 provides purely for backward compatibility.

Figure 3.1 – 3.2: New Virtual Machine Wizard — Win10A and Win10B Summaries

Both wizard summaries confirm identical Generation 2 configuration, 4096 MB memory, and the same ISO source, with each VM's hard disk stored under its own named folder — keeping the two machines fully independent while ensuring their baseline configuration is consistent for a fair comparison later, when Win10A and Win10B are used to demonstrate two different Storage Spaces resiliency models.





3.2 — Operating System Installation

With both VMs provisioned, Windows Setup was run against the shared ISO. The **Custom: Install Windows only (advanced)** option was selected over the Upgrade path — appropriate here since both VMs were installing onto entirely new, unformatted virtual disks with no prior Windows installation to preserve. Regional settings (language, time and currency format, and keyboard layout) were configured to English (South Africa) on both machines during setup, so that locale-dependent behaviour — date formats, currency symbols, and keyboard input — would match the environment both machines were built to represent.

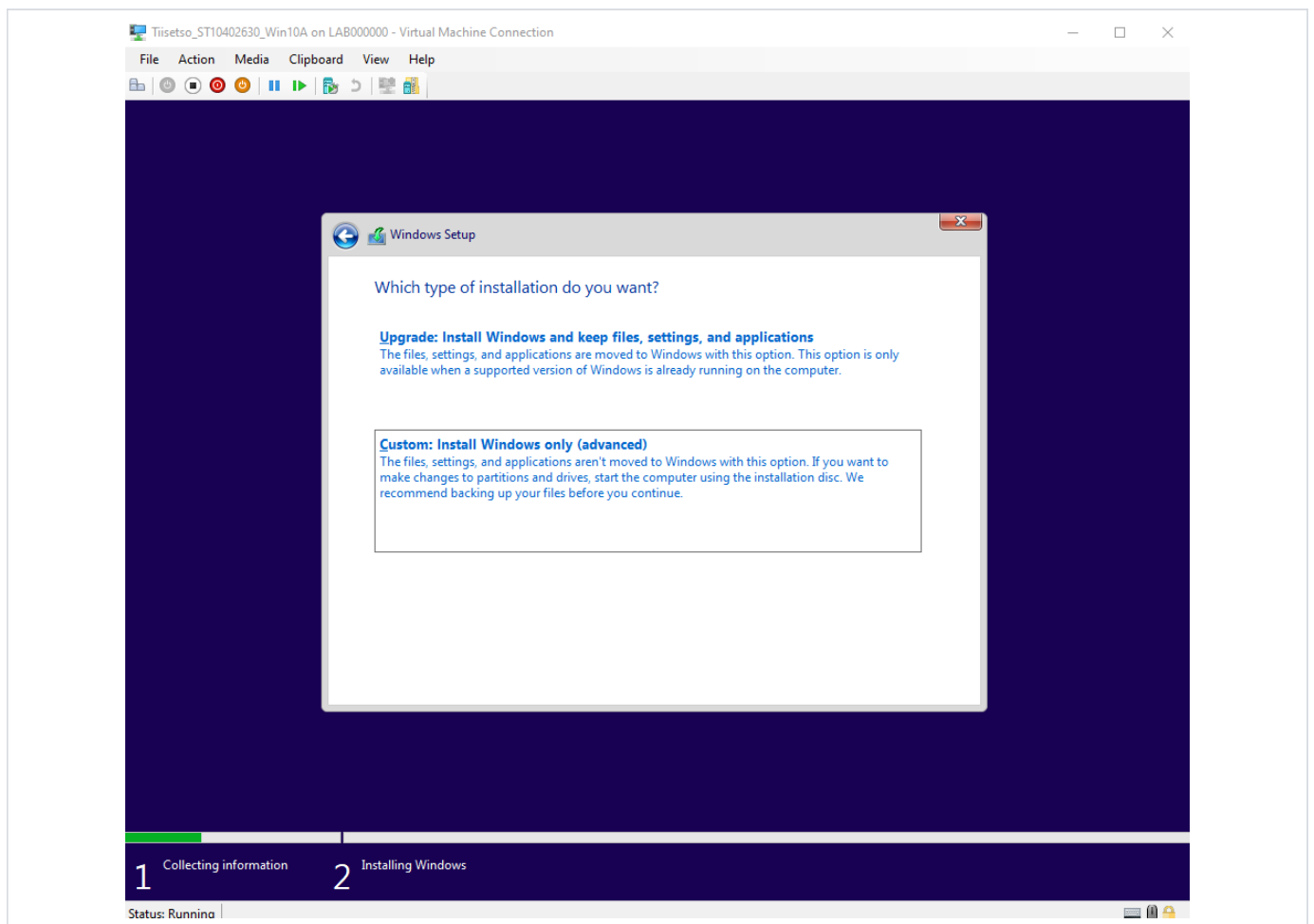
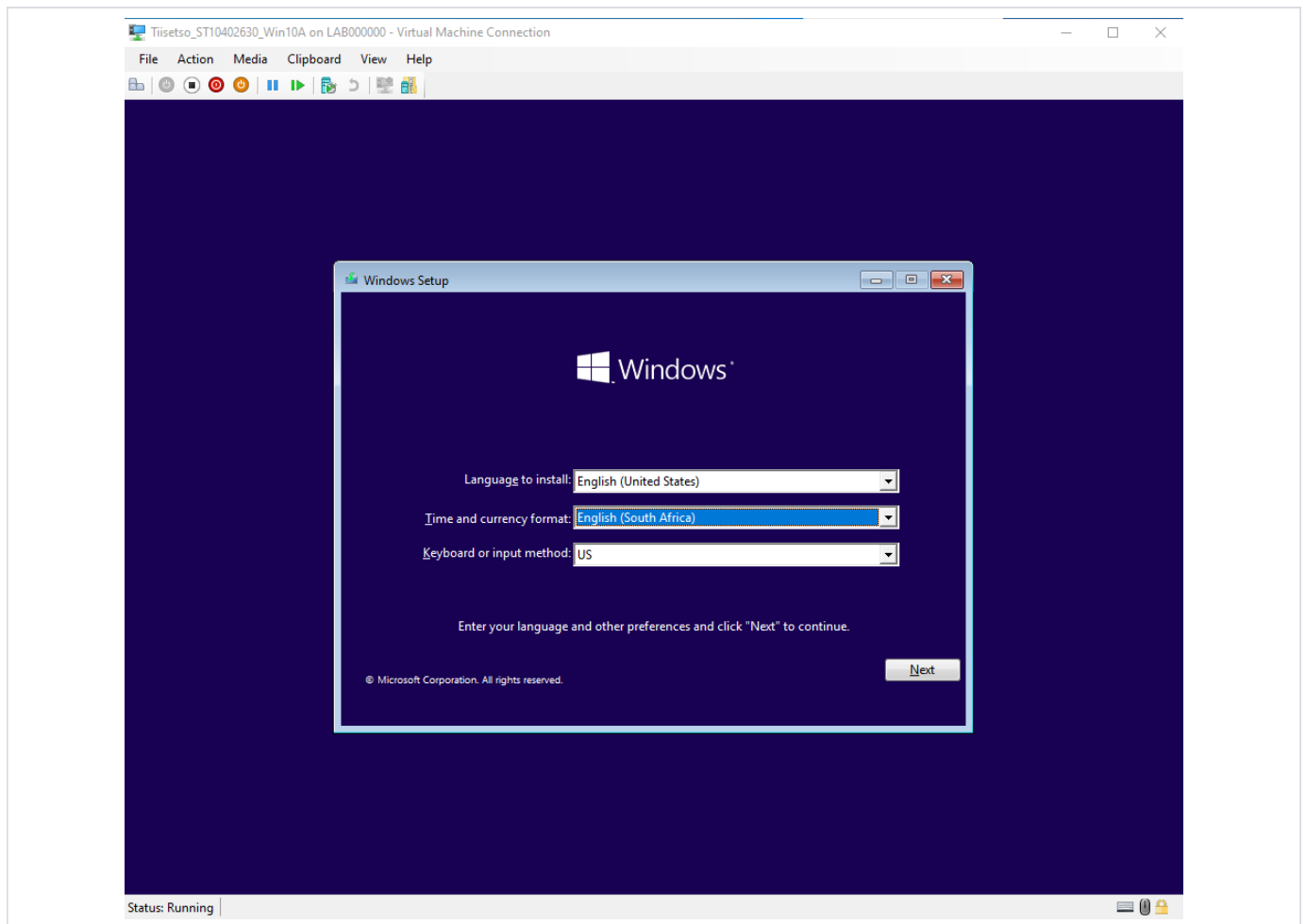
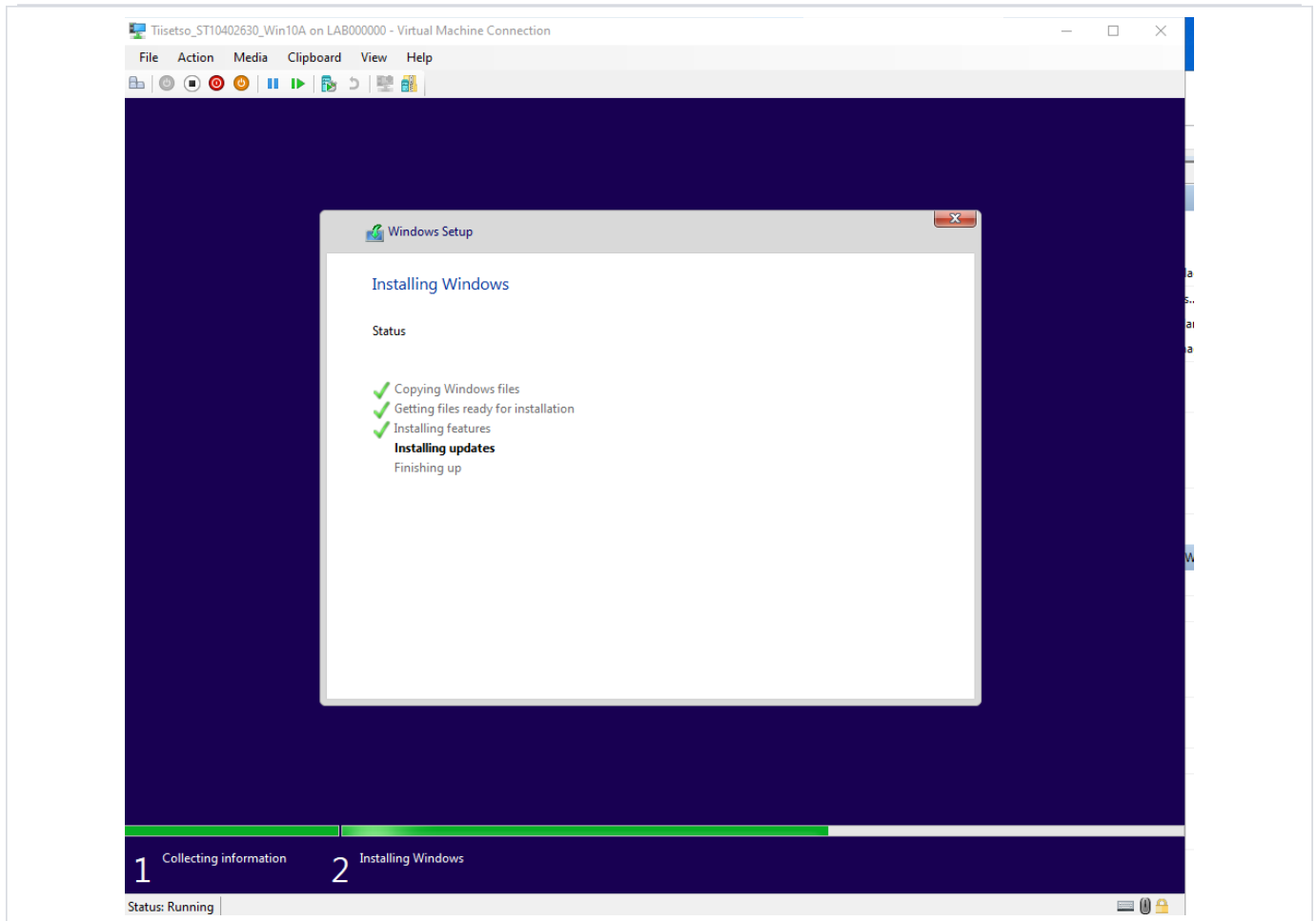
Figure 3.3: Custom Install Selected

Figure 3.4: Regional Settings Configured to South Africa**Figure 3.5: Installation Completed**

Setup progressed through file copy, feature installation, and update installation before reaching the finishing stage — confirming a clean, uninterrupted install on both machines with no driver or feature failures reported.



3.3 — Device Identification

Once each installation reached the desktop, both machines were renamed from their default auto-generated device name to a naming convention that identifies the candidate and the machine's role — Tiisetso-Win10A and Tiisetso-Win10B — verified via **Settings** → **System** → **About**. Consistent, identifiable device naming is a basic but important administrative discipline: in any environment with more than one machine, an auto-generated name (e.g. DESKTOP-8F3K2L1) gives an administrator no information at a glance, whereas a naming convention tied to the machine's owner or role makes asset tracking, remote troubleshooting, and inventory audits considerably faster.

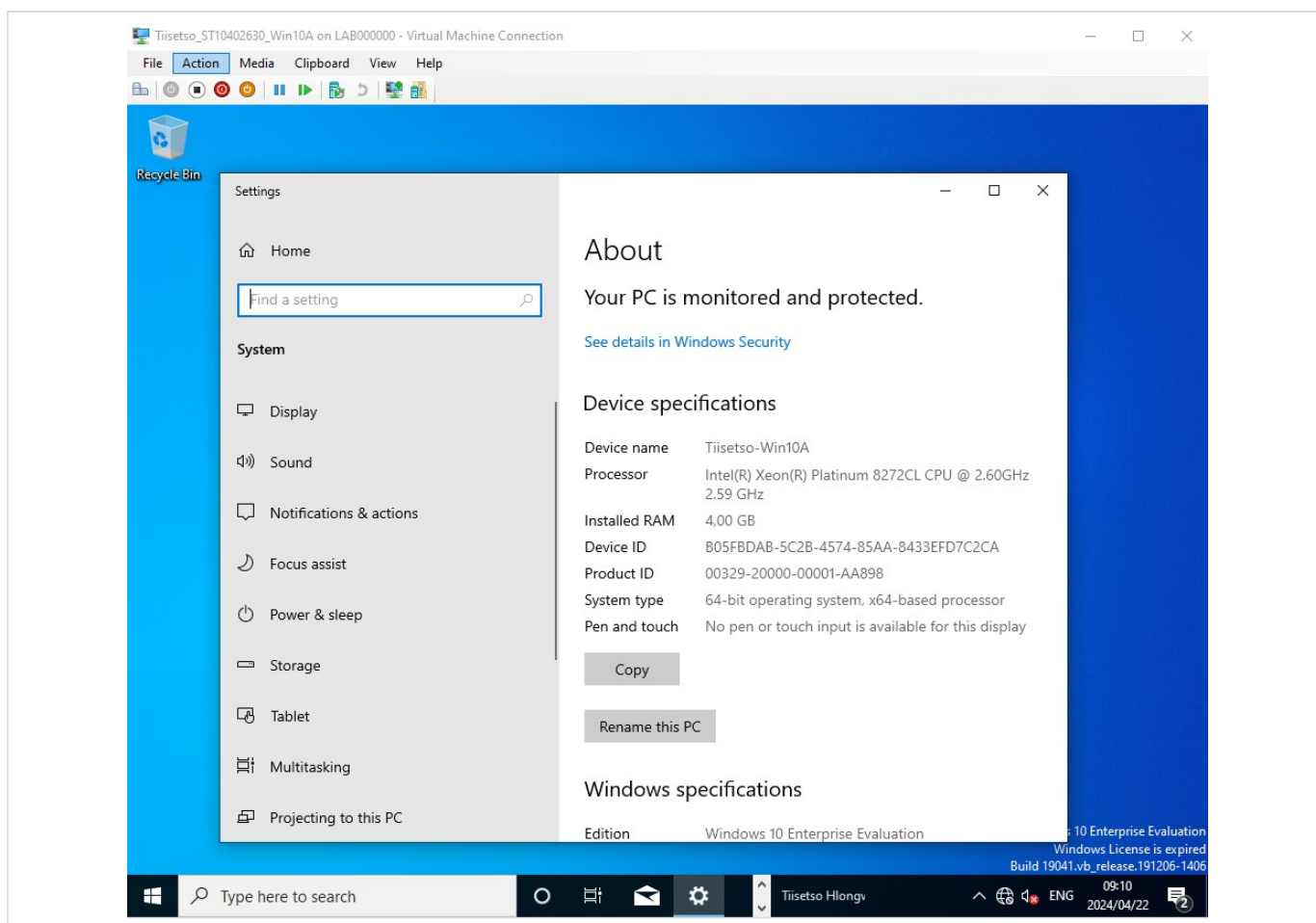
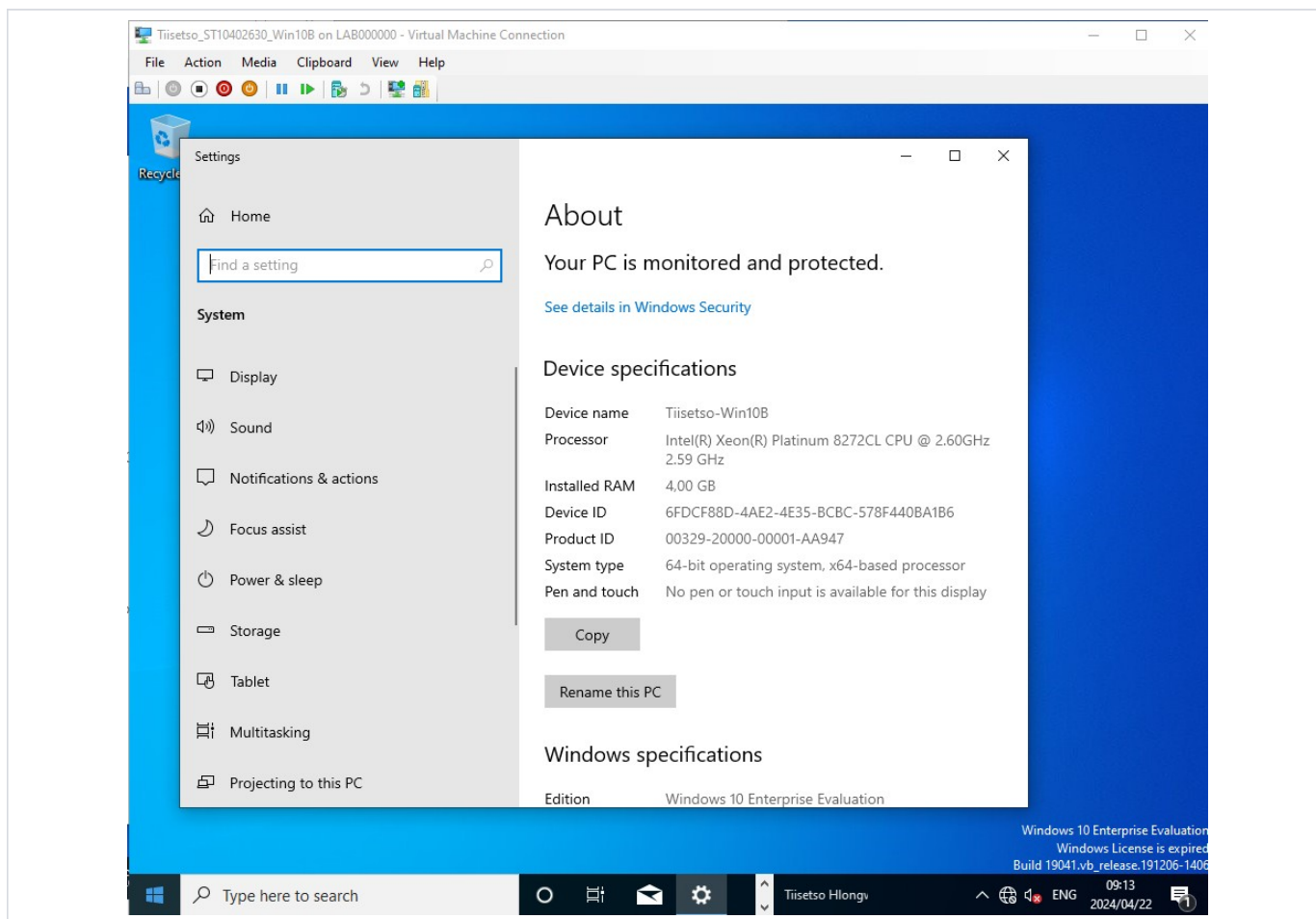
Figure 3.6: Device Renamed — Tiisetso-Win10A

Figure 3.7: Device Renamed — Tiisetso-Win10B

04

SYSTEM UTILITIES & TROUBLESHOOTING

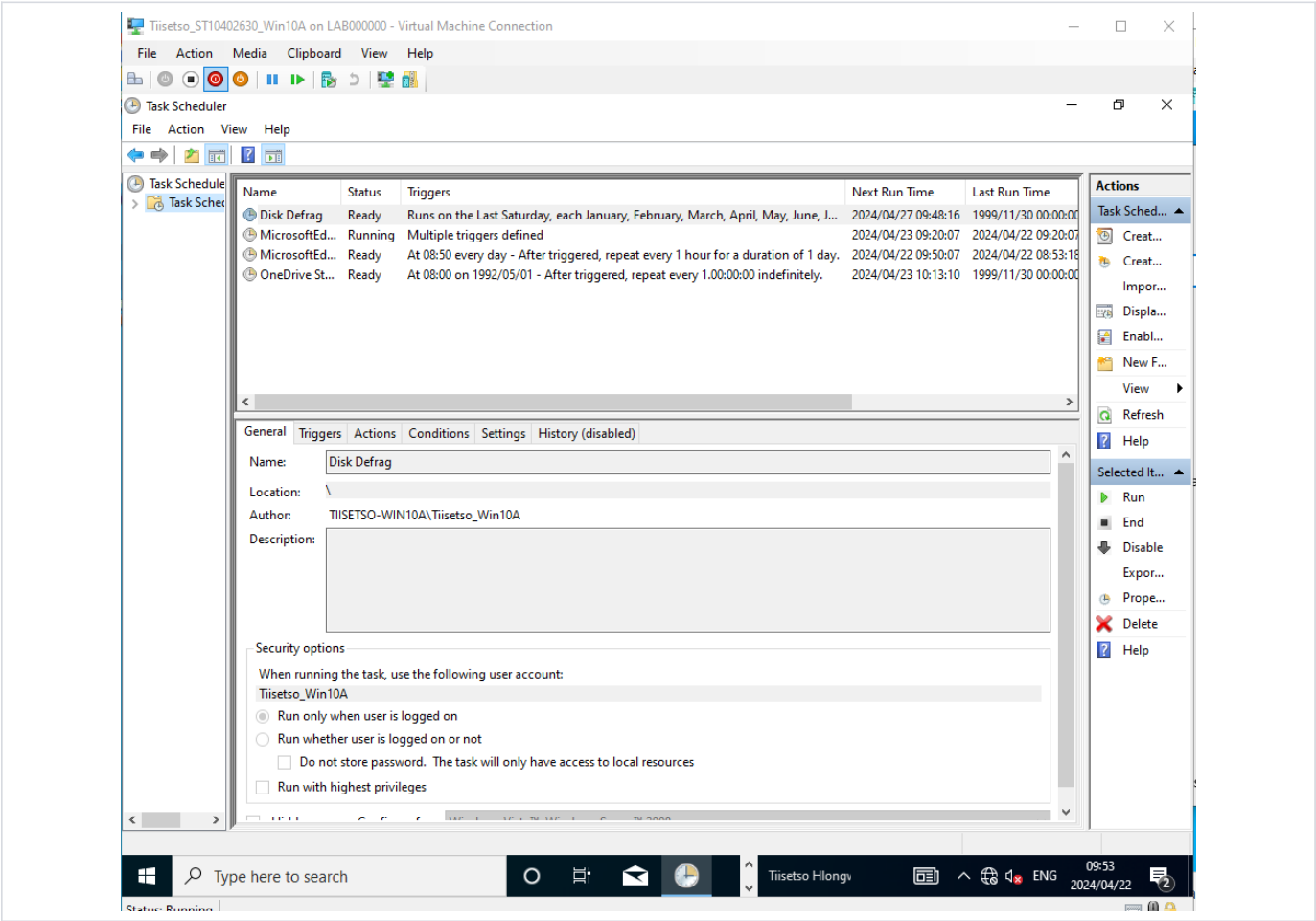
With both machines installed, three native Windows 10 utilities were exercised on `Tiisetso_Win10A` — Task Scheduler, Driver Verifier Manager, and Device Manager — each representing a different category of day-to-day administrative work: automated maintenance, driver integrity auditing, and hardware-level troubleshooting.

4.1 — Automated Maintenance via Task Scheduler

A scheduled task was configured to run **Disk Defragmentation** automatically on the last Saturday of each month, rather than relying on a user to remember to run it manually. Monthly was chosen as a reasonable cadence for an endpoint under normal use — frequent enough to keep file fragmentation from degrading read performance on a traditional or lightly-used disk, without running a maintenance task so often that it competes unnecessarily with foreground workloads. The task's security context was scoped to run only when the user is logged on, which avoids the disk-maintenance job attempting to run against a locked or powered-off session.

Figure 4.1: Disk Defragmentation Scheduled — Monthly Trigger

The Task Scheduler library confirms the Disk Defrag task with a **Ready** status, a trigger of "Runs on the last Saturday, each January, February... ", and its next scheduled run time populated correctly — confirming the trigger was accepted and is active rather than merely configured.



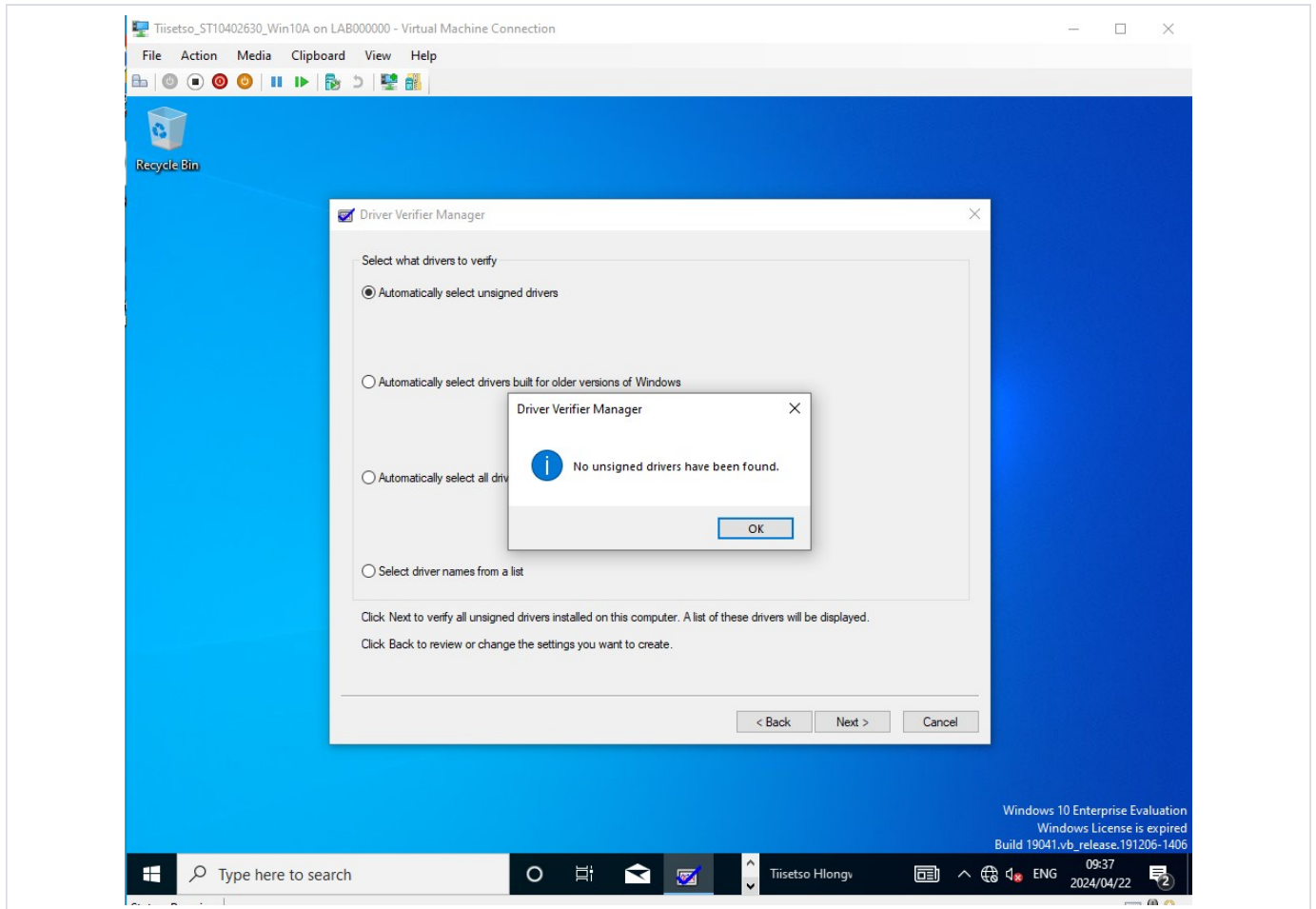
4.2 — Driver Signature Auditing via Driver Verifier Manager

Driver Verifier Manager was run with **Automatically select unsigned drivers** to audit every installed driver on the machine against Microsoft's code-signing requirement. Unsigned or improperly signed drivers are one of

the more common vectors for both instability (a poorly-tested driver crashing the kernel) and malicious code (a driver running with kernel-level privilege bypassing normal application sandboxing), so this check is a legitimate first-line integrity audit before trusting a freshly-installed machine.

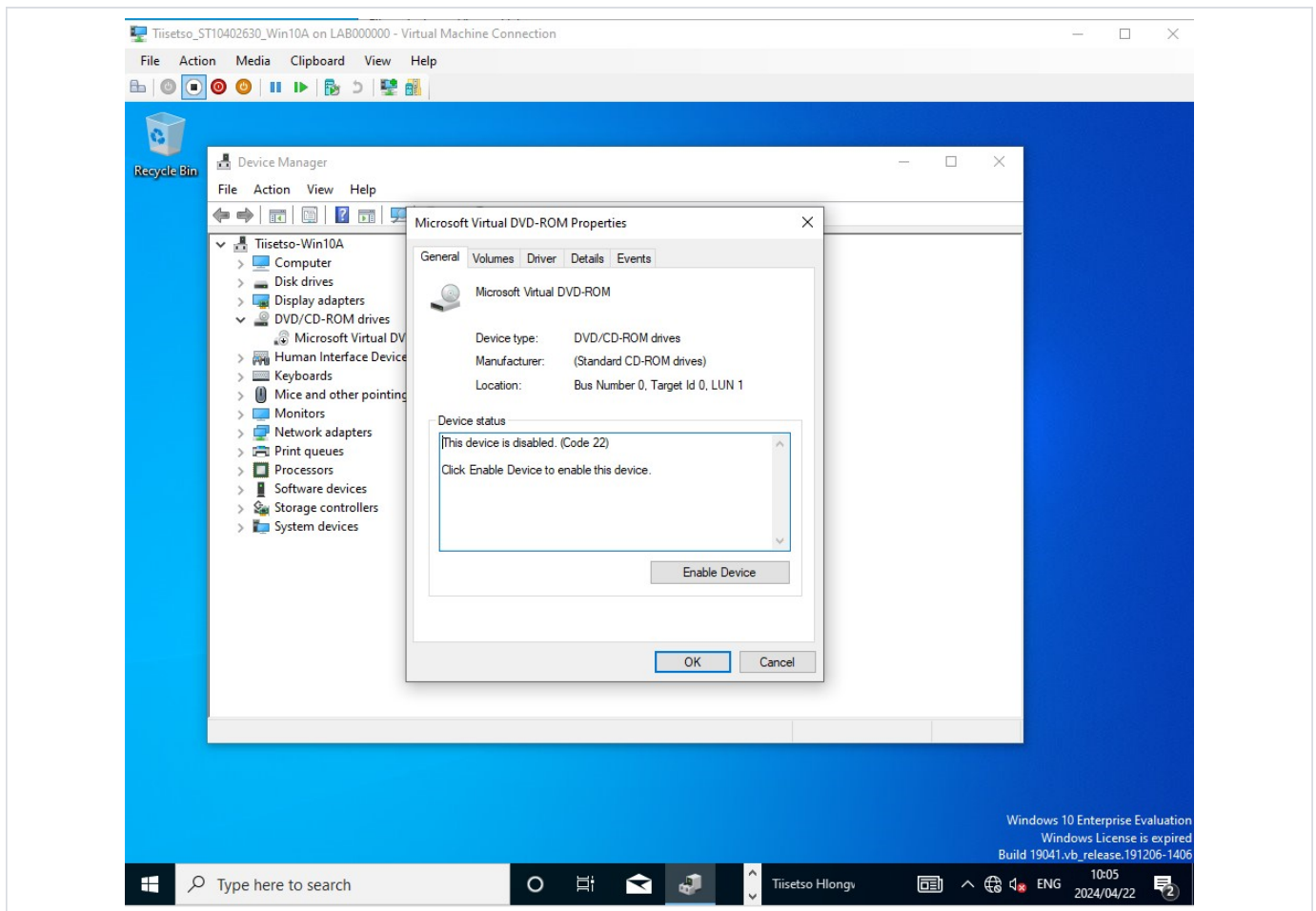
Figure 4.2: No Unsigned Drivers Found

The scan returned "**No unsigned drivers have been found**", confirming that every driver on this installation is properly signed — an expected and reassuring result on a clean VM image, and the baseline that ongoing driver installations should be checked against.



4.3 — Hardware Troubleshooting via Device Manager

The Microsoft Virtual DVD-ROM device was intentionally disabled through Device Manager to demonstrate the diagnostic workflow an administrator would follow when a piece of hardware is reported as non-functional — the **Device status** panel showing "This device is disabled. (Code 22)" is exactly the information Device Manager surfaces when a technician needs to confirm whether a hardware fault is a physical failure or, as here, simply a disabled device requiring re-enablement rather than a driver reinstall or hardware replacement.

Figure 4.3: DVD-ROM Controller Disabled (Code 22)

05

STORAGE & DISK MANAGEMENT

Both client machines were used to demonstrate **Storage Spaces**, Windows' software-defined storage resiliency feature — deliberately configured with two **different** resiliency types across Win10A and Win10B, so that the trade-off between them could be shown directly rather than described only in theory. Disk Management volume verification and the underlying Hyper-V virtual hard disk provisioning are also documented, since Storage Spaces sits on top of that lower storage layer.

5.1 — Storage Spaces: Parity Resiliency (Win10B)

On Win10B, a storage pool (TiisetsoTestData, drive **E:**) was created with **Parity** resiliency and a 200 GB maximum size against a 252 GB pool. Parity resiliency writes data alongside parity information across the pool's drives — similar in principle to RAID-5 — which protects against a single drive failure while using storage more efficiently than a full mirror, at the cost of requiring at least three physical drives and slower write performance than a mirrored space. This makes parity a reasonable choice for a volume expected to hold data that is written occasionally but read frequently, such as the department test data this space represents.

Figure 5.1: Storage Pool Created — Parity Resiliency

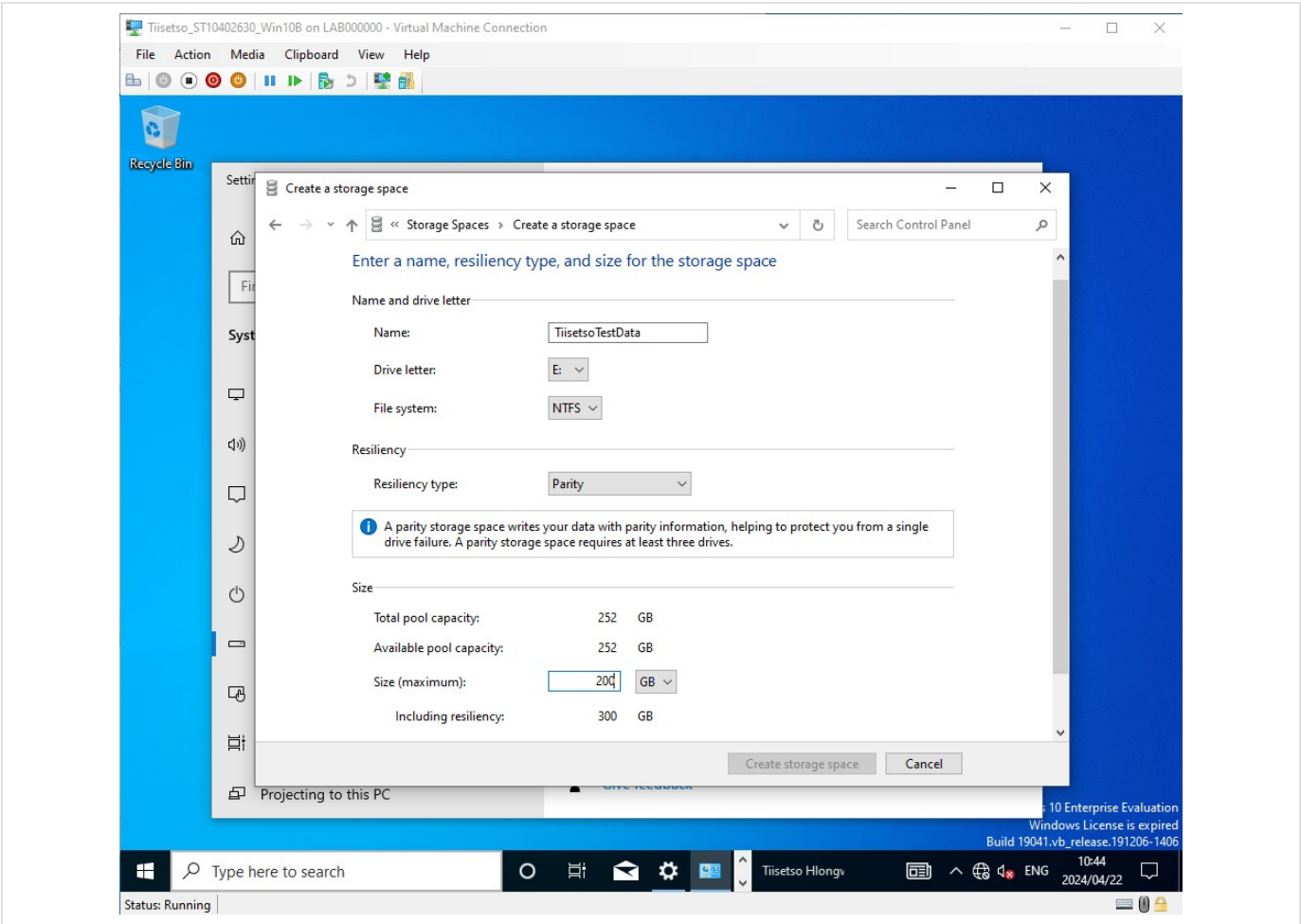
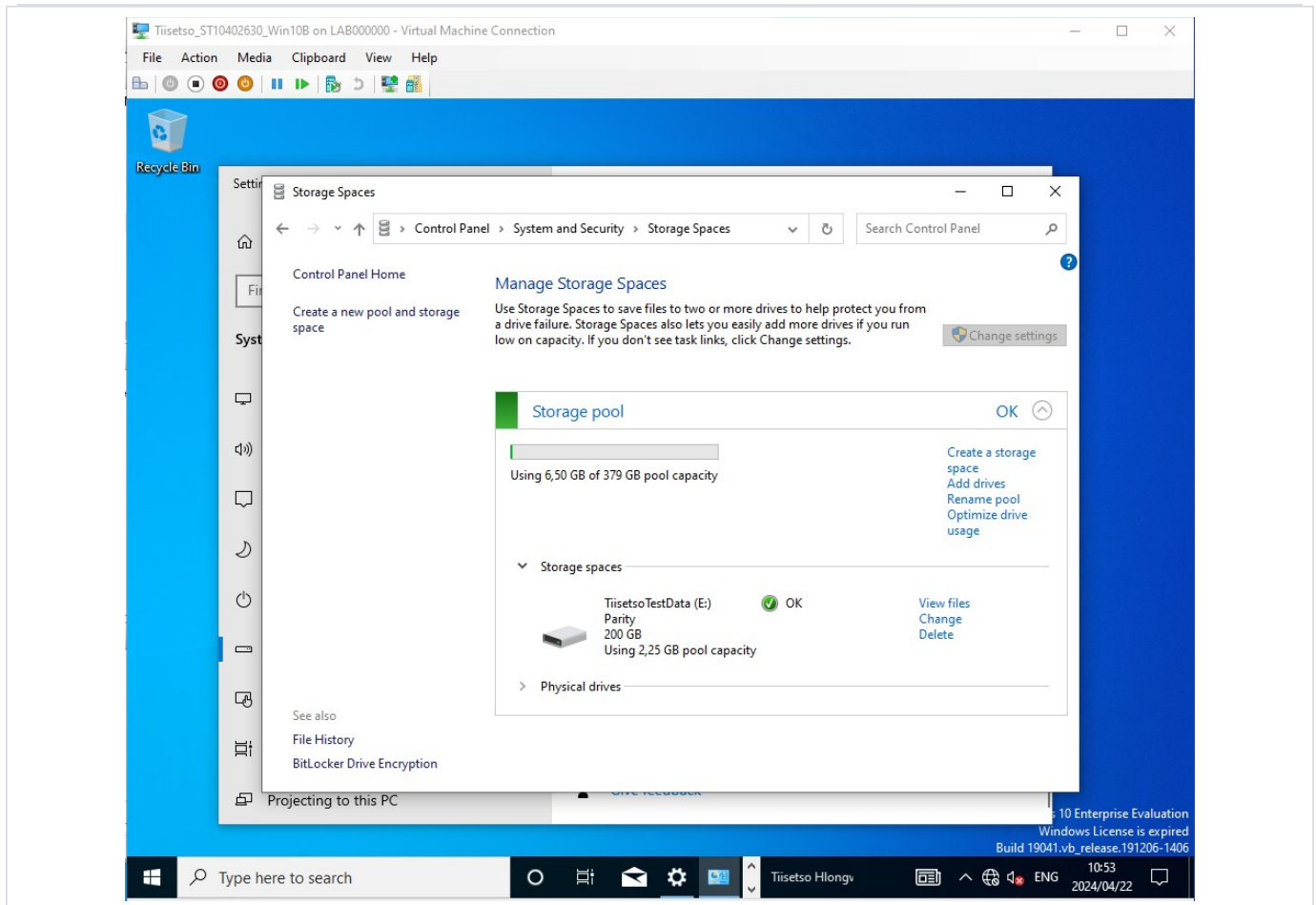


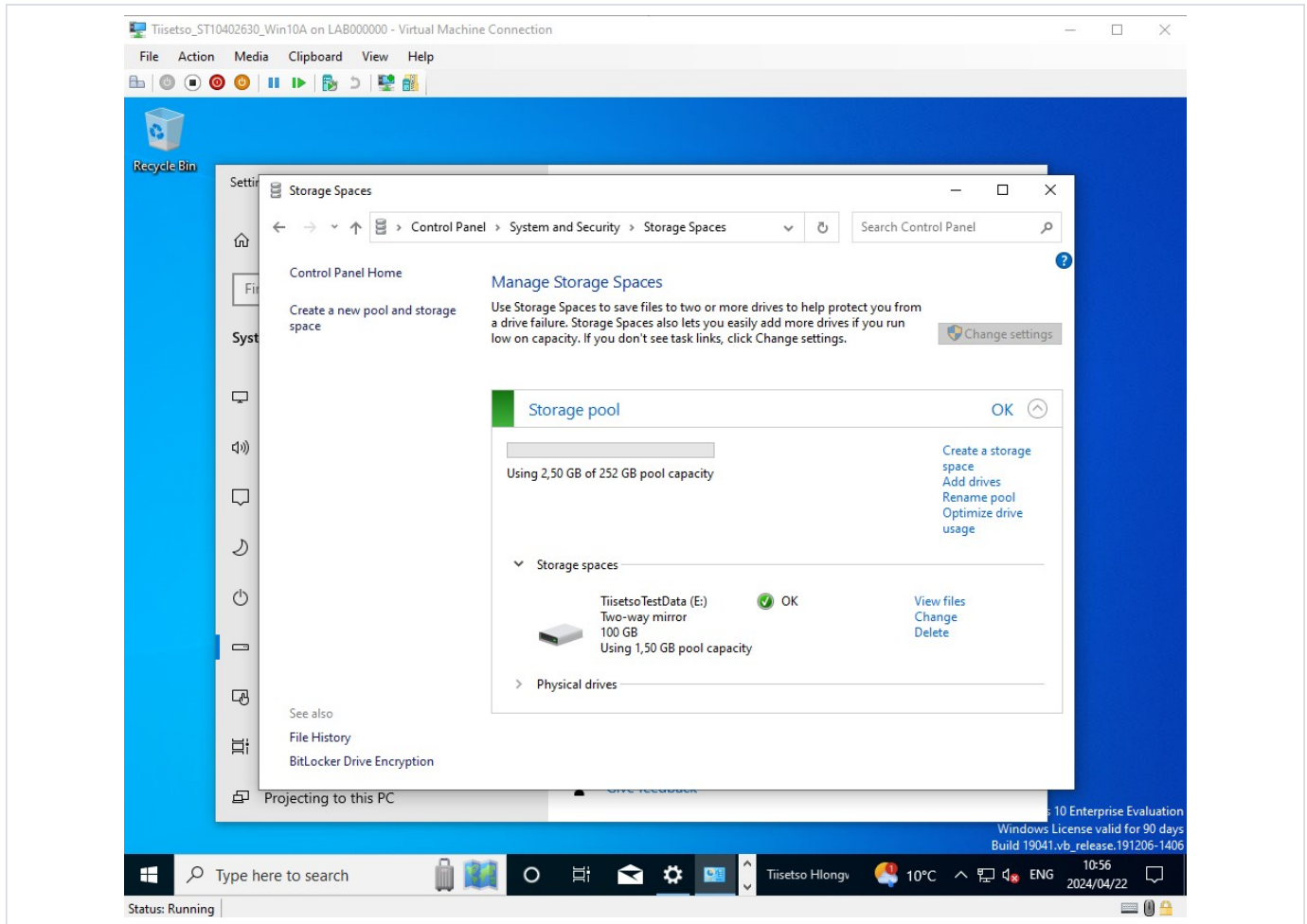
Figure 5.2: Parity Storage Space Verified on Win10B

The completed pool shows a healthy **OK** status on the TiisetsoTestData (E:) space, confirming the 200 GB parity volume was created successfully and is using 2.25 GB of pool capacity.



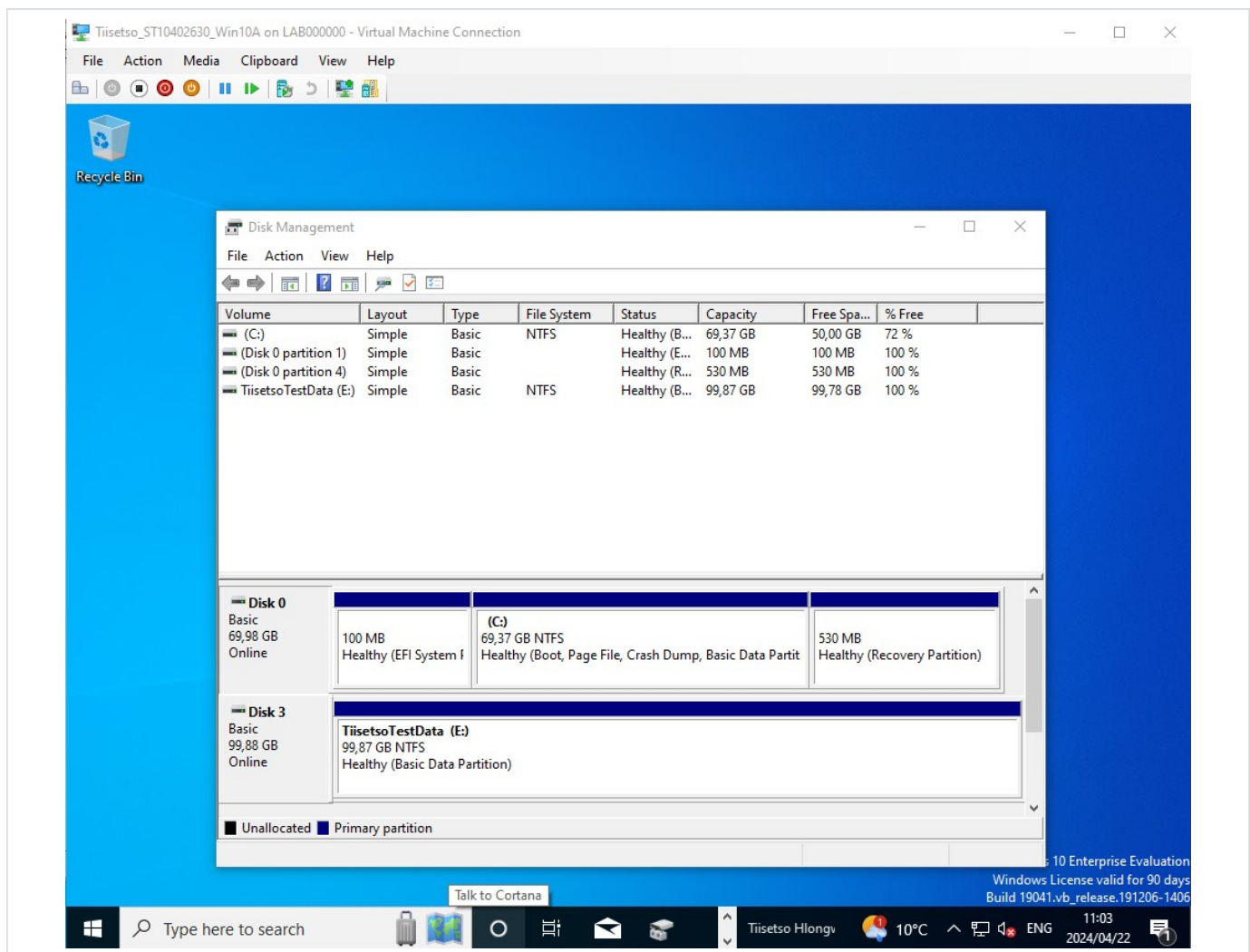
5.2 — Storage Spaces: Two-Way Mirror Resiliency (Win10A)

On Win10A, the same `TiisetsoTestData` naming convention was used, but this time configured with **two-way mirror** resiliency (100 GB, against a 252 GB pool). A two-way mirror writes every block of data to two separate physical locations rather than computing parity, which trades storage efficiency (roughly half the pool's raw capacity becomes usable, versus parity's better ratio) for **faster write performance and a simpler recovery path** — a single drive can fail with an immediate, direct copy of the data still available on the surviving drive, with no parity recalculation required. This is the more appropriate resiliency choice for a volume expected to be written to frequently, illustrating why the resiliency type is a deliberate design decision rather than a default left unchanged.

Figure 5.3: Two-Way Mirror Storage Space Verified on Win10A

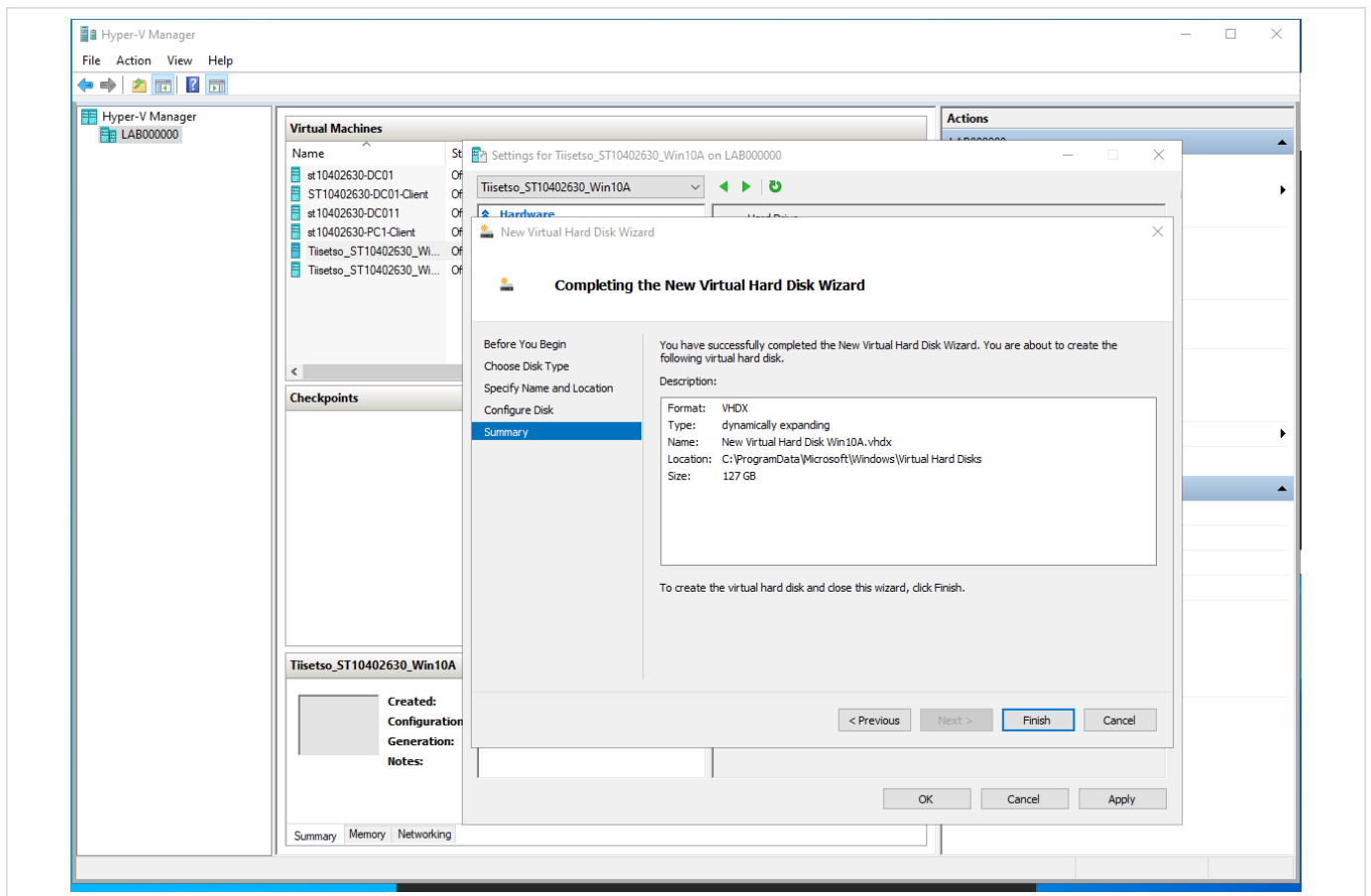
5.3 — Disk Volume Verification

Disk Management on Win10A confirms the resulting disk layout: the system disk (Disk 0, holding the EFI system partition, the boot/OS partition, and the recovery partition) alongside Disk 3, which hosts the TiisetsoTestData (E:) basic data partition created by the Storage Spaces wizard. All volumes report a **Healthy** status, confirming the storage configuration is not just created but functioning correctly at the file-system level.

Figure 5.4: Disk Management — Volumes Confirmed Healthy

5.4 — Virtual Hard Disk Provisioning (Hyper-V Layer)

Beneath the guest OS's own disk management, the underlying **dynamically expanding VHDX** virtual hard disks were provisioned through Hyper-V Manager for both Win10A and Win10B. A dynamically expanding disk was chosen over a fixed-size disk because it allocates host storage on demand as the guest writes data, rather than reserving the full disk size upfront — a sensible trade-off for a lab environment where predictable performance matters less than conserving host disk space across multiple VMs.

Figure 5.5: Hyper-V Virtual Hard Disk Wizard — Win10A

06

USERS & GROUP MANAGEMENT

This section covers local identity administration on `Tiisetso_Win10B`, carried out through **Computer Management** → **Local Users and Groups**: individual user accounts were created first, department-aligned security groups were then created and populated with those users, and privilege was finally assigned at the group level rather than to individual accounts.

6.1 — User Account Provisioning

Individual local user accounts were created through the **New User** dialog, each with a full name, an initial password, and **User cannot change password** enabled so that account credentials remain under administrative control rather than being alterable by the end user — an appropriate restriction for standard accounts in a managed lab environment, where password policy is intended to be governed centrally through Local Group Policy (covered in Section 07) rather than left to individual discretion.

Figure 6.1: Creating a New Local User

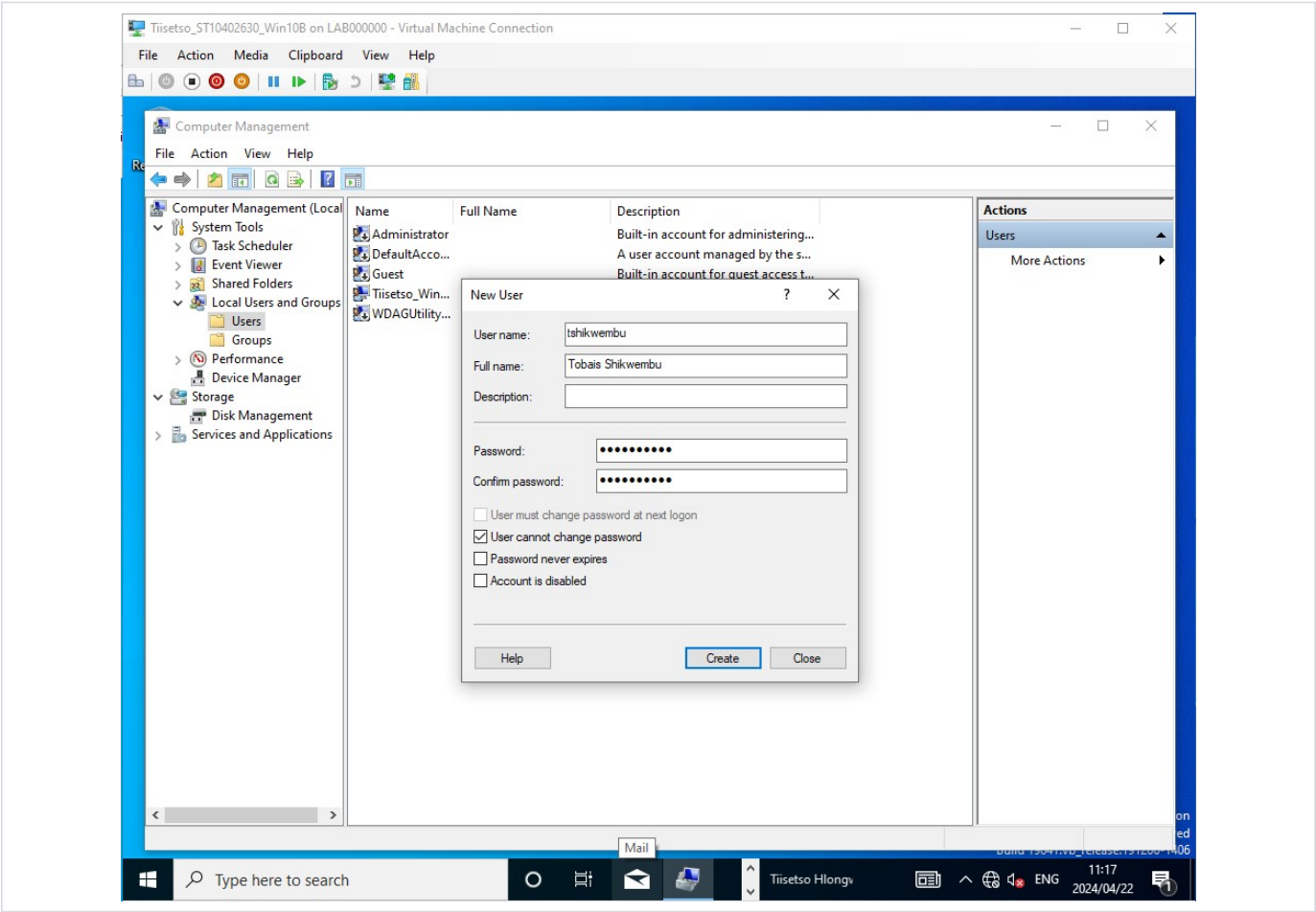
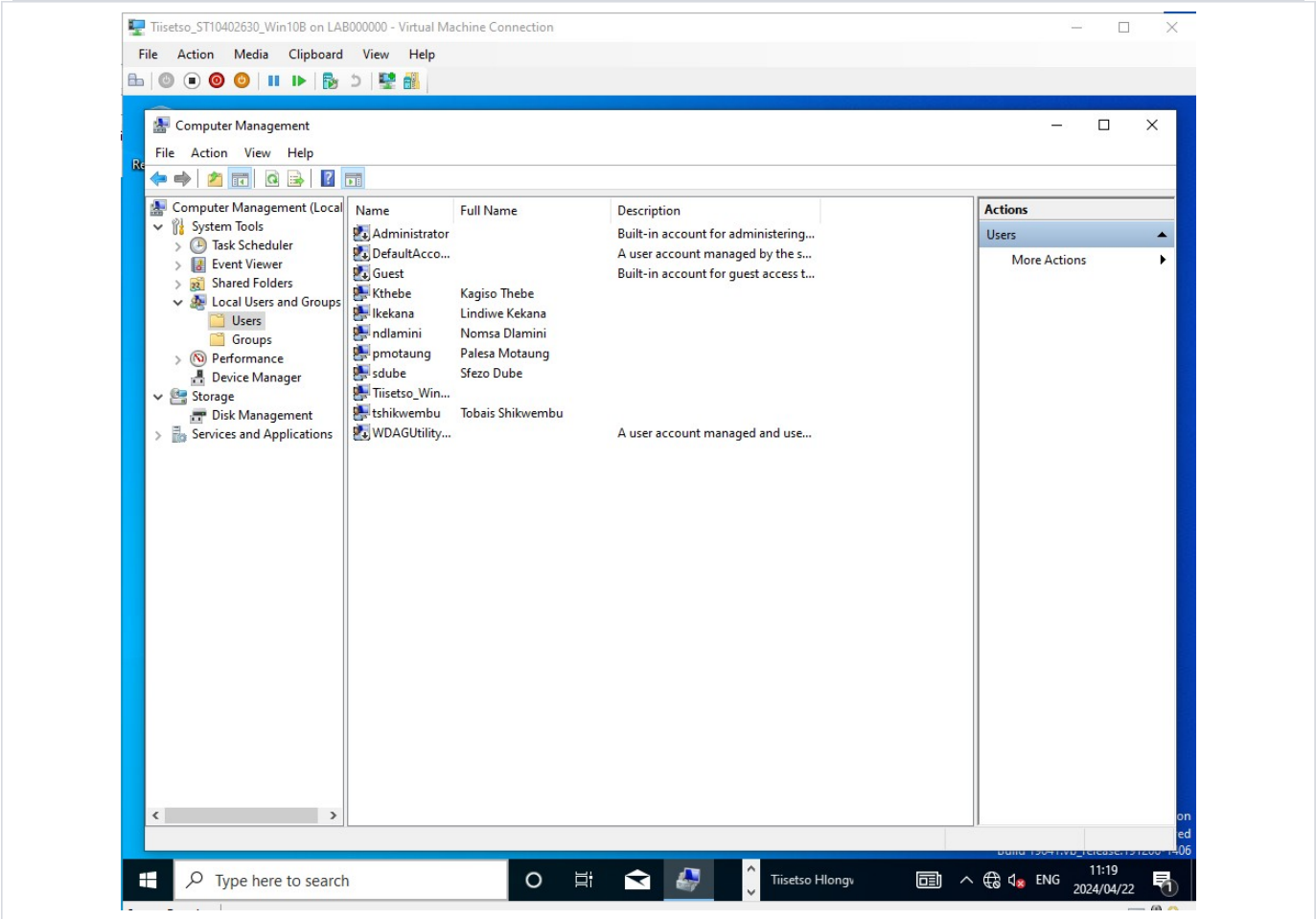


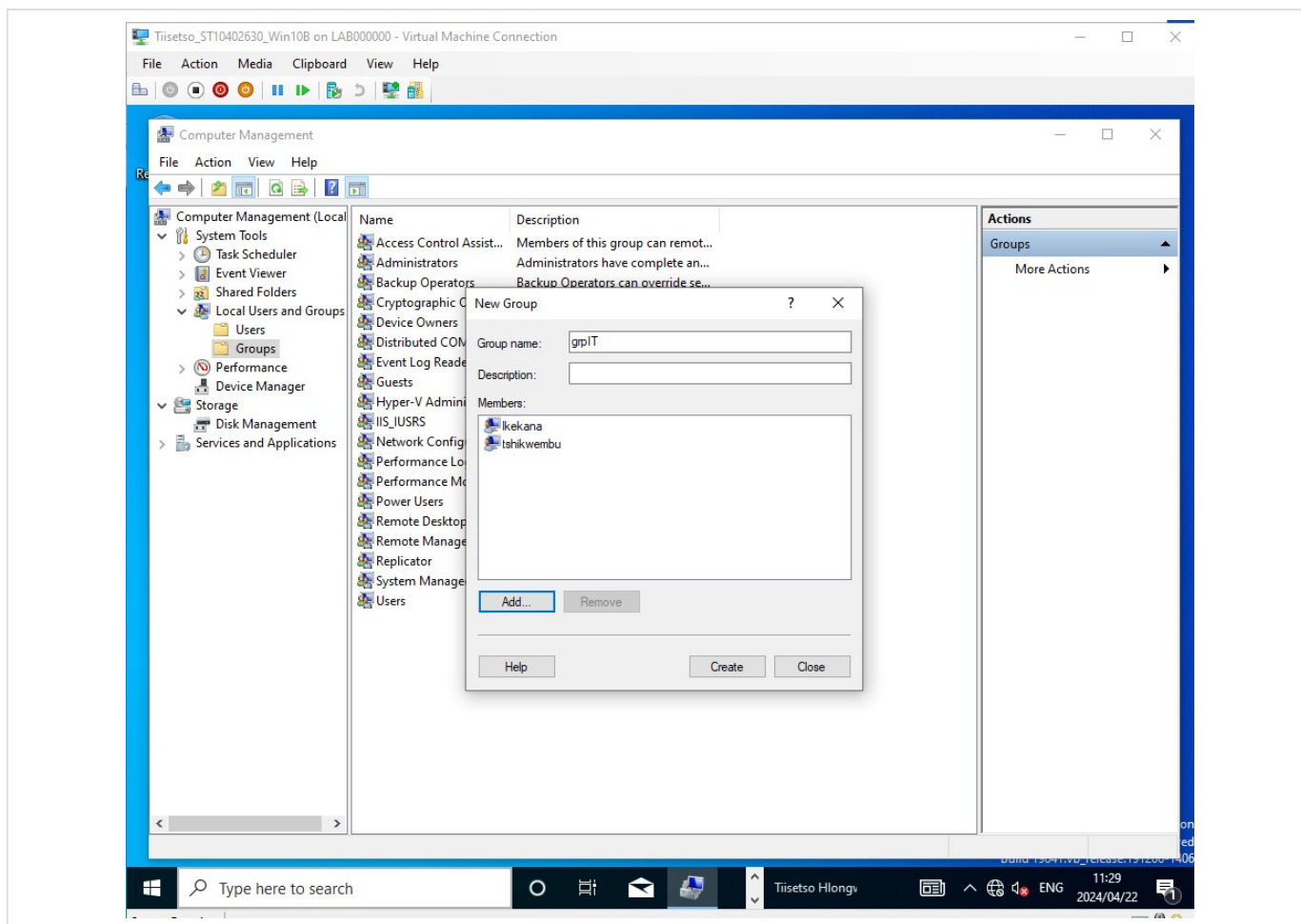
Figure 6.2: All User Accounts Created

The completed Users list confirms five standard accounts alongside the built-in `Administrator`, `DefaultAccount`, and `Guest` accounts — each standard account showing its full name against its logon name, ready to be assigned to a department group.

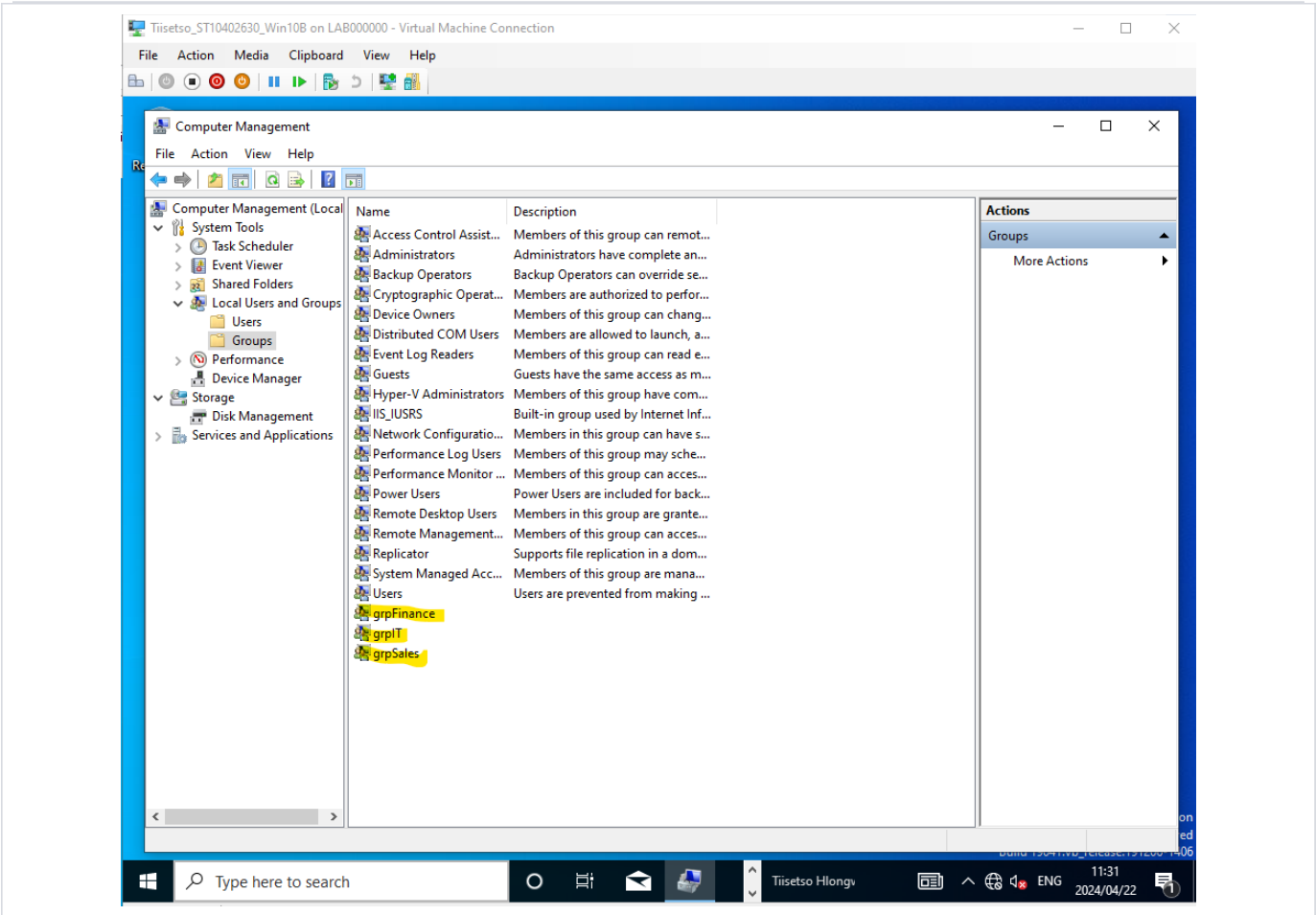


6.2 — Security Group Creation

Rather than managing permissions per individual user, department-scoped local security groups were created — grpIT, grpFinance, and grpSales — with the relevant user accounts added as members at creation time. Grouping users by department **before** assigning any privilege is the standard access-control pattern: it means a future change in departmental membership (someone joining or leaving IT, for example) only requires updating group membership once, rather than hunting down and reapplying every individual permission that person had been granted directly.

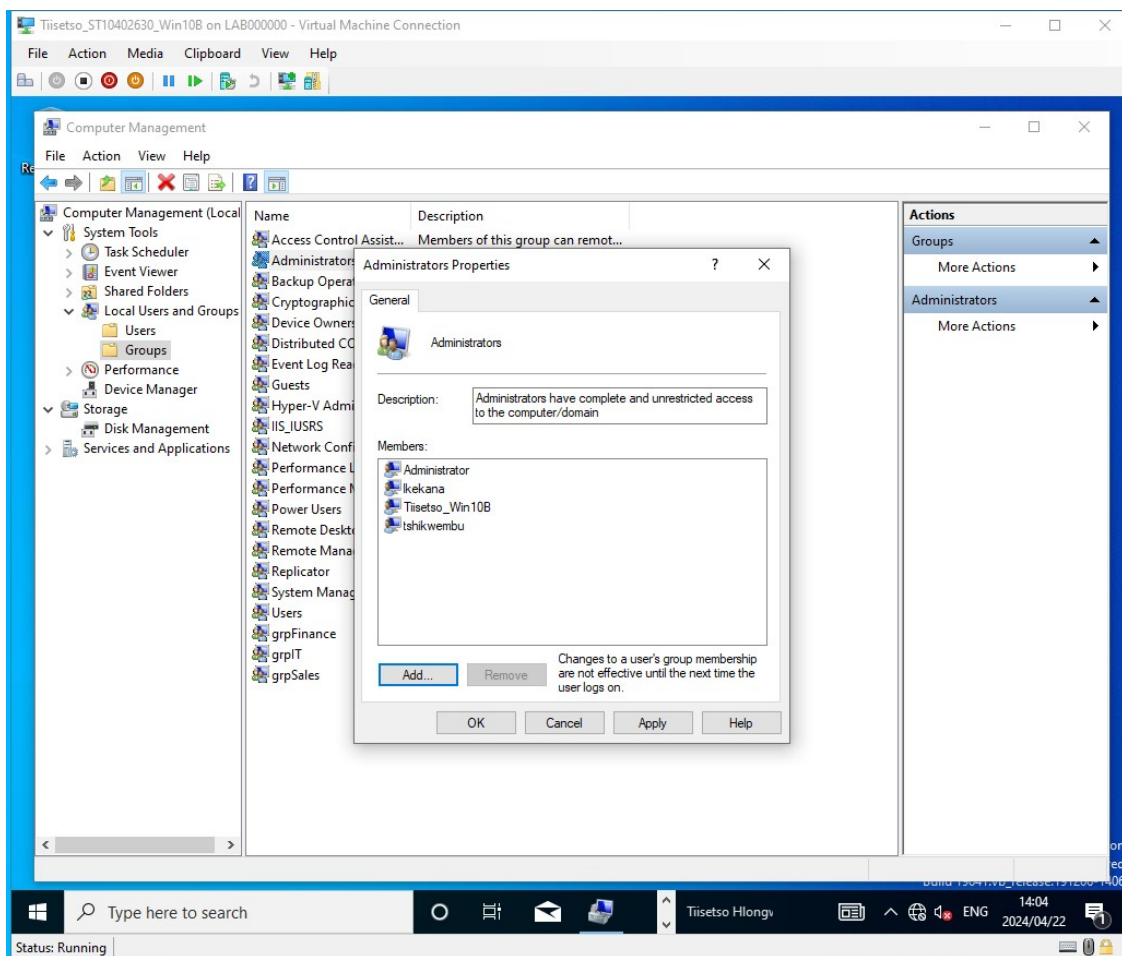
Figure 6.3: Creating the grpIT Security Group and Adding Members**Figure 6.4: All Groups Created**

The Groups list confirms grpFinance, grpIT, and grpSales now exist alongside the built-in Windows groups, each ready to be referenced for privilege assignment.



6.3 — Privilege Assignment

With the IT department's user accounts grouped under grpIT, that group — not the individual accounts — was added as a member of the built-in local **Administrators** group. This grants full administrative rights to every current and future member of grpIT without ever touching the Administrators group membership again: adding a new IT staff member later is a one-step change (add them to grpIT), rather than a two-step change that risks being forgotten (add them to grpIT, then remember to also grant Administrator rights directly). This is the same group-based privilege model used at domain scale in enterprise Active Directory environments, demonstrated here at the local level.

Figure 6.5: grpIT Added to the Local Administrators Group

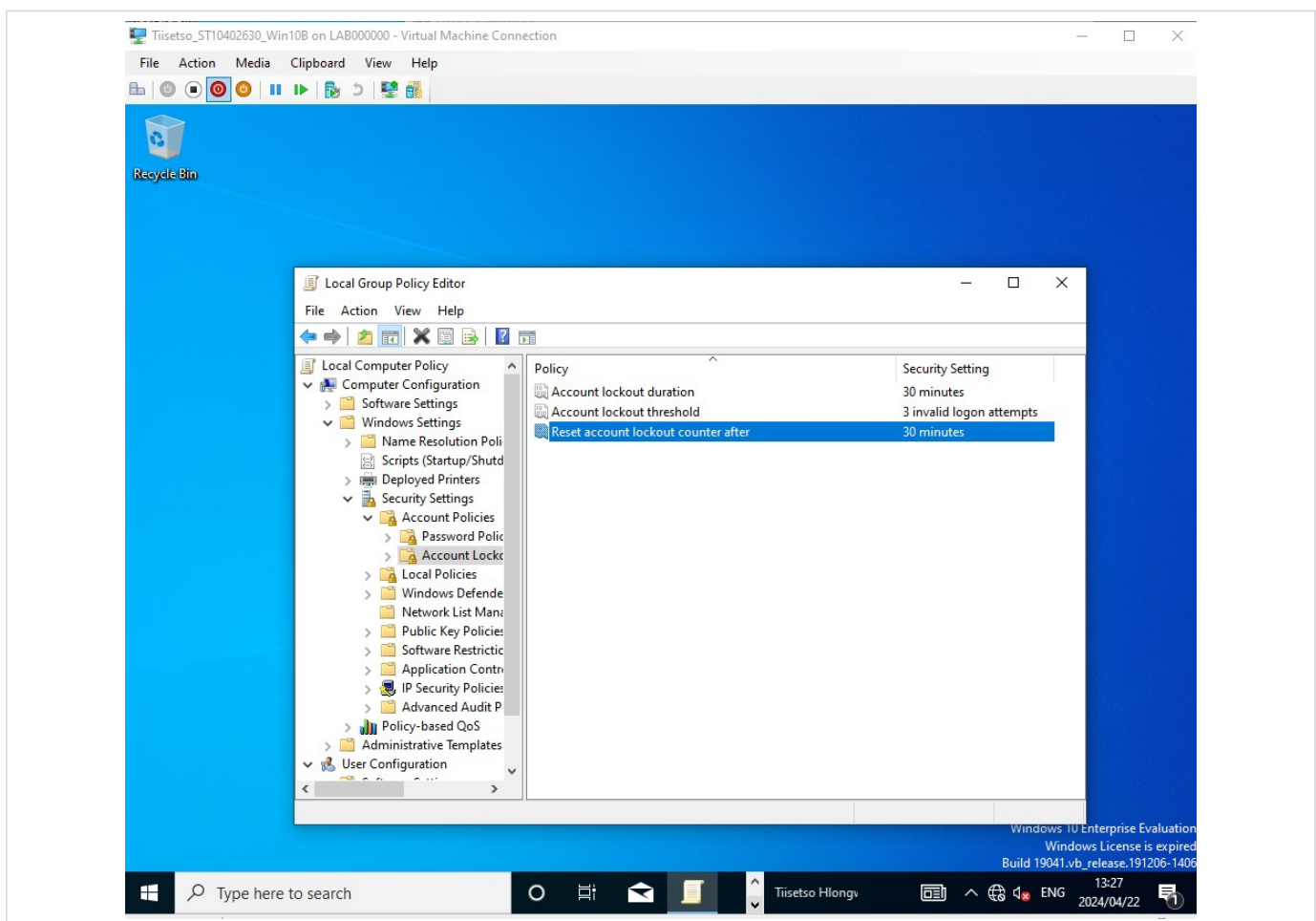
07 SECURITY POLICY CONFIGURATION

A local security baseline was enforced on `Tiisetso_Win10B` through the **Local Group Policy Editor**, covering account lockout behaviour, password strength, User Account Control, and interactive logon protections. Each setting below was chosen deliberately rather than left at its Windows default, and the section closes with the policy verified against a live Windows Update check and a full system restart.

7.1 — Account Lockout Policy

Account lockout was configured with a threshold of **3 invalid logon attempts** before an account locks, with the lockout counter reset window set to **30 minutes**. A low threshold of 3 attempts meaningfully slows a brute-force or password-guessing attack against a local account, since an attacker gains only three guesses before being locked out for half an hour — a substantial deterrent against automated credential attacks without being so aggressive that a legitimate user mistyping their password twice risks losing access.

Figure 7.1: Account Lockout Policy Configured



7.2 — Password Policy

Password must meet complexity requirements was enabled, enforcing a mix of character types (uppercase, lowercase, digits, and symbols) in every password set on the machine. This was combined with a wider password policy: a minimum length of 8 characters, a password history of 15 remembered passwords to prevent immediate reuse, and a maximum password age of 29 days to force periodic rotation. Complexity and minimum length together raise the practical difficulty of both brute-force and dictionary-based password

attacks, while the remembered-password history stops a user from defeating the rotation requirement by simply cycling back to a previous, already-compromised password.

Figure 7.2: Password Complexity Requirement Enabled

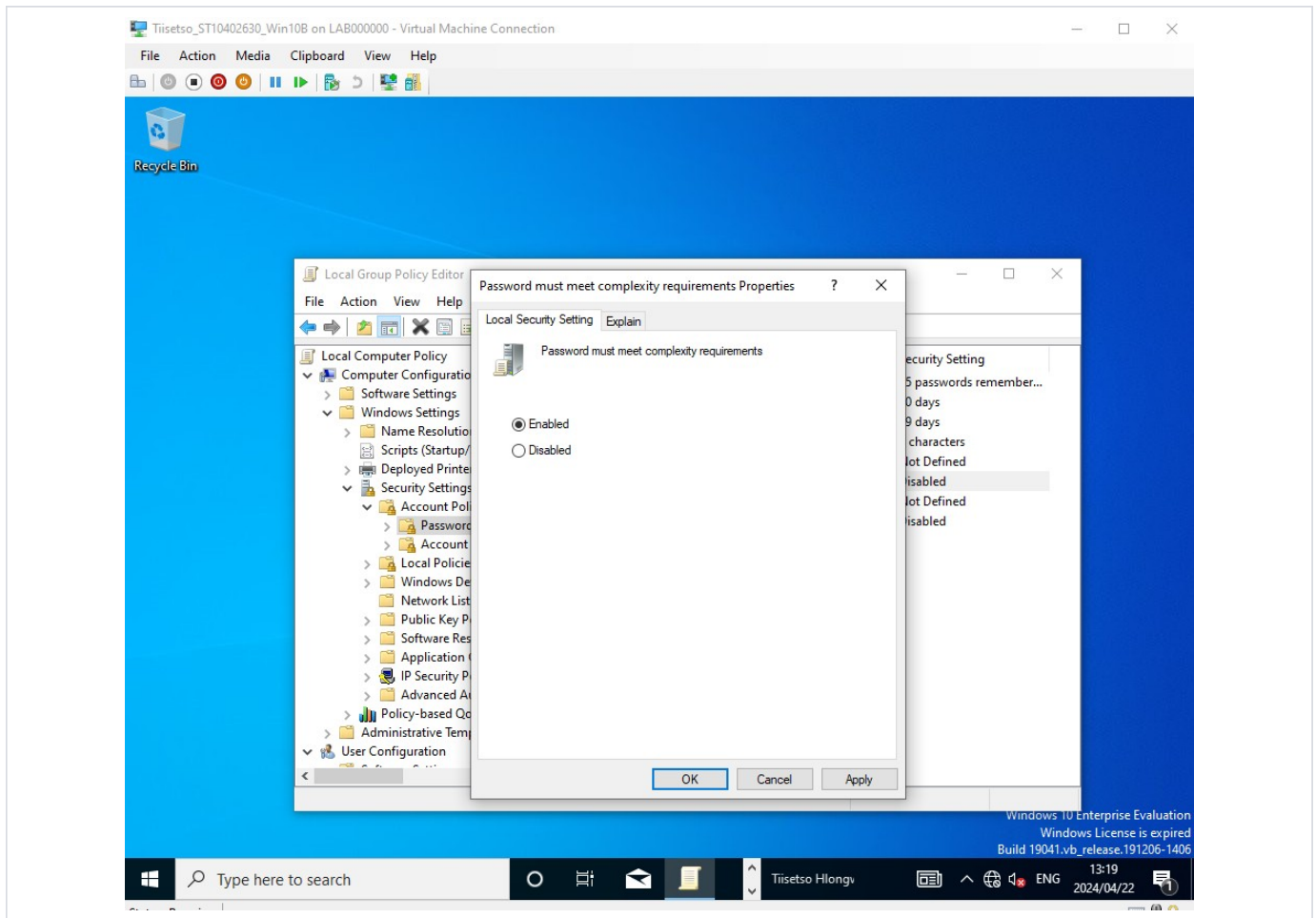
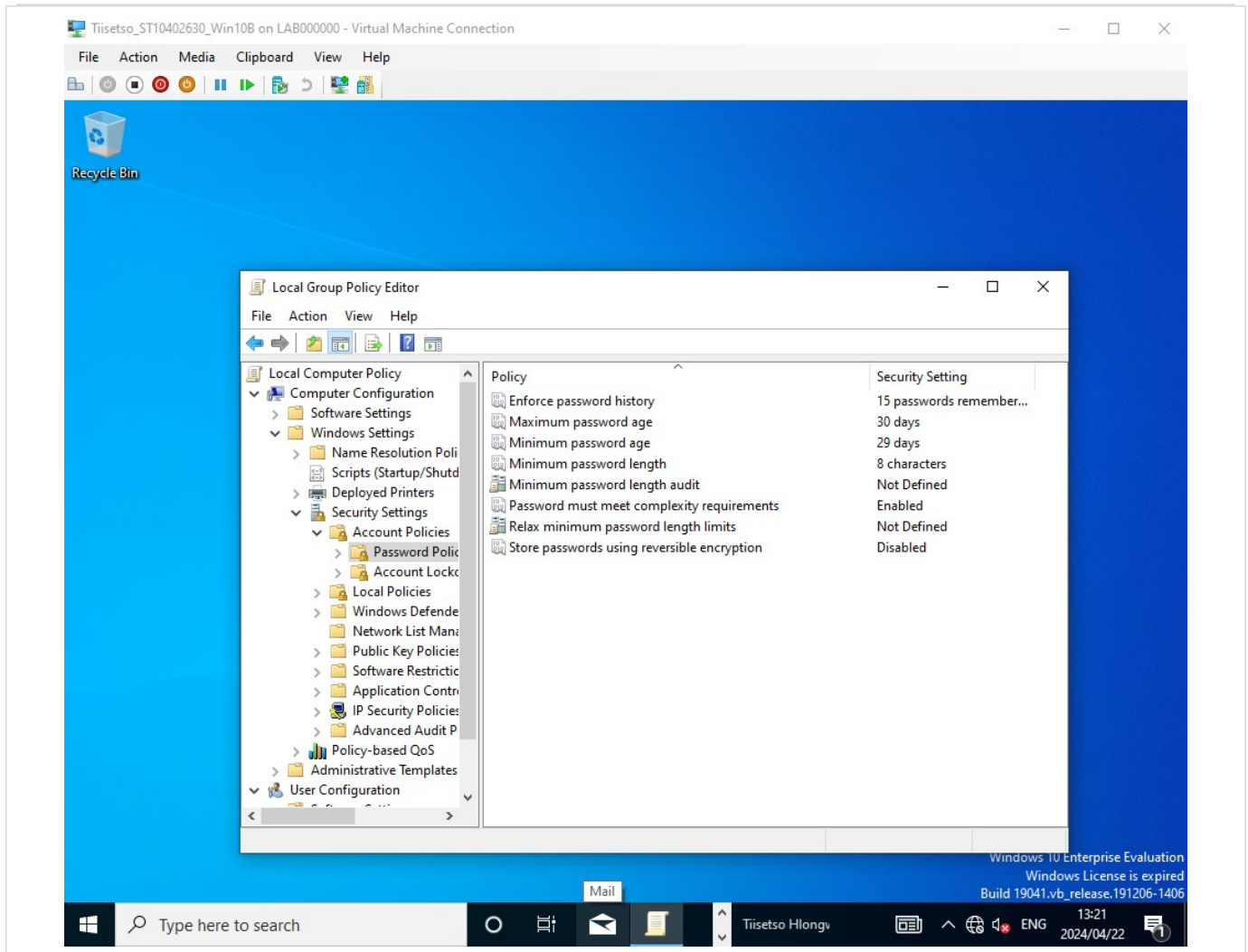


Figure 7.3: Full Password Policy Configured

The Password Policy node confirms all settings applied together: 15 remembered passwords, a 29-day maximum age, an 8-character minimum length, complexity enabled, and reversible encryption left disabled — since there was no application dependency requiring it, and enabling it would only weaken password storage security for no operational benefit.



7.3 — Interactive Logon Banner

An **Acceptable Computer Usage Policy** banner was configured to display at every logon attempt, combining a message title ("ACCEPTABLE COMPUTER USAGE POLICY") with detailed message text covering prohibited actions — no downloading or installing of illegal software, no viewing of inappropriate material, no eating or drinking at the workstation, no connecting unauthorised external devices, and a requirement that the PC remain locked when unattended. A logon banner serves a genuine compliance purpose beyond its technical configuration: it creates a documented record that every user acknowledged the organisation's acceptable-use policy before being granted access, which matters both for enforcement and for any later disciplinary or legal process.

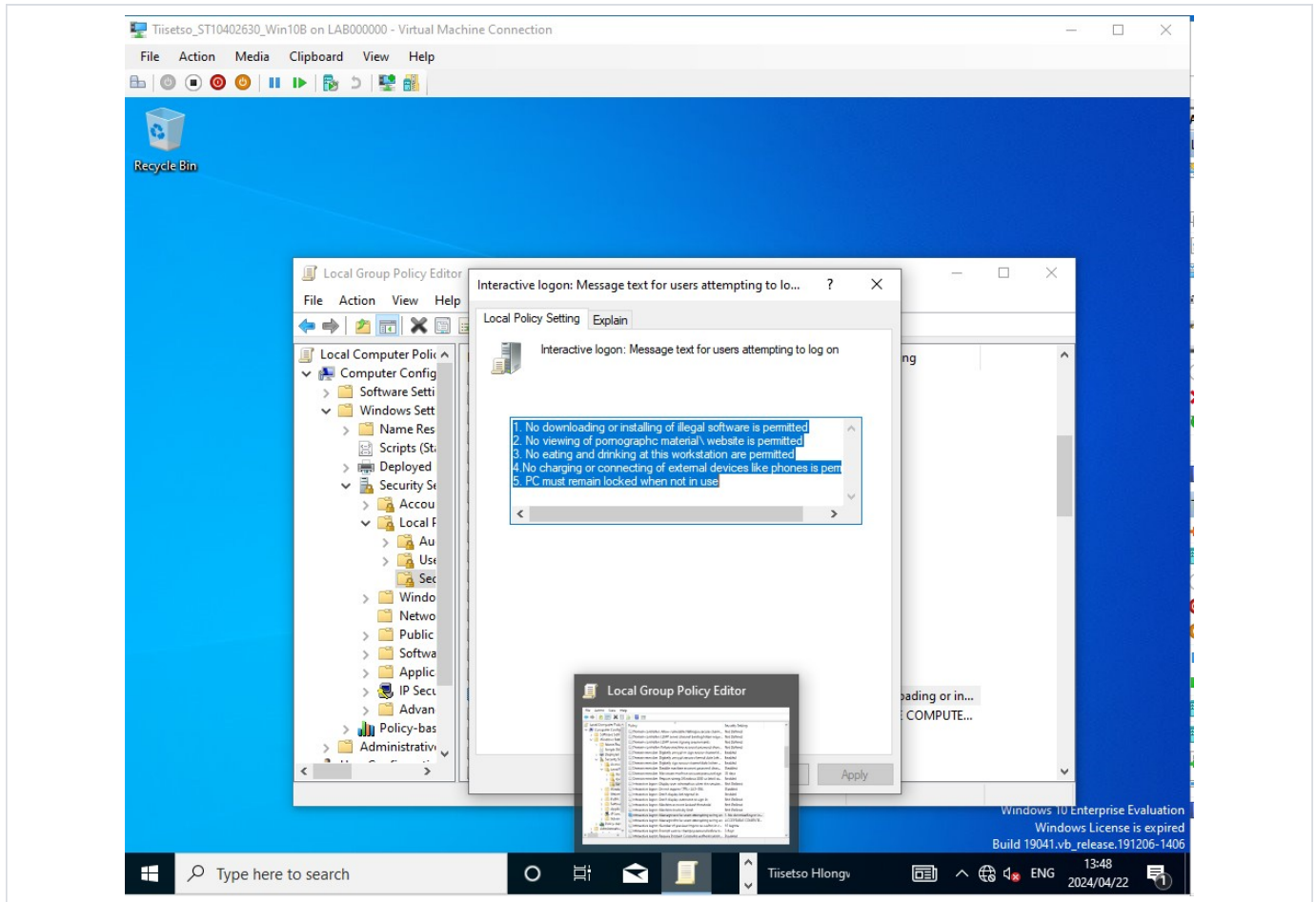
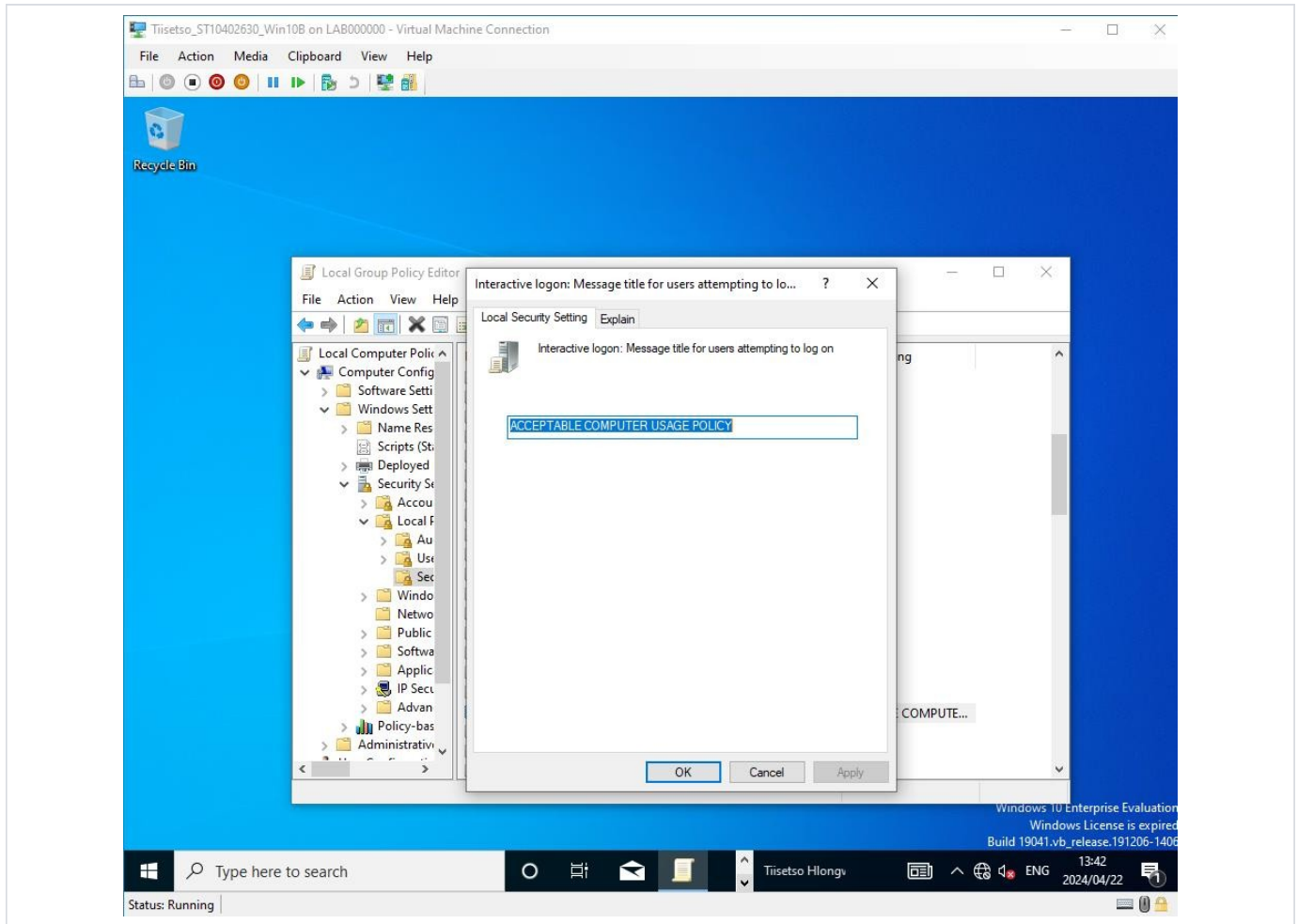
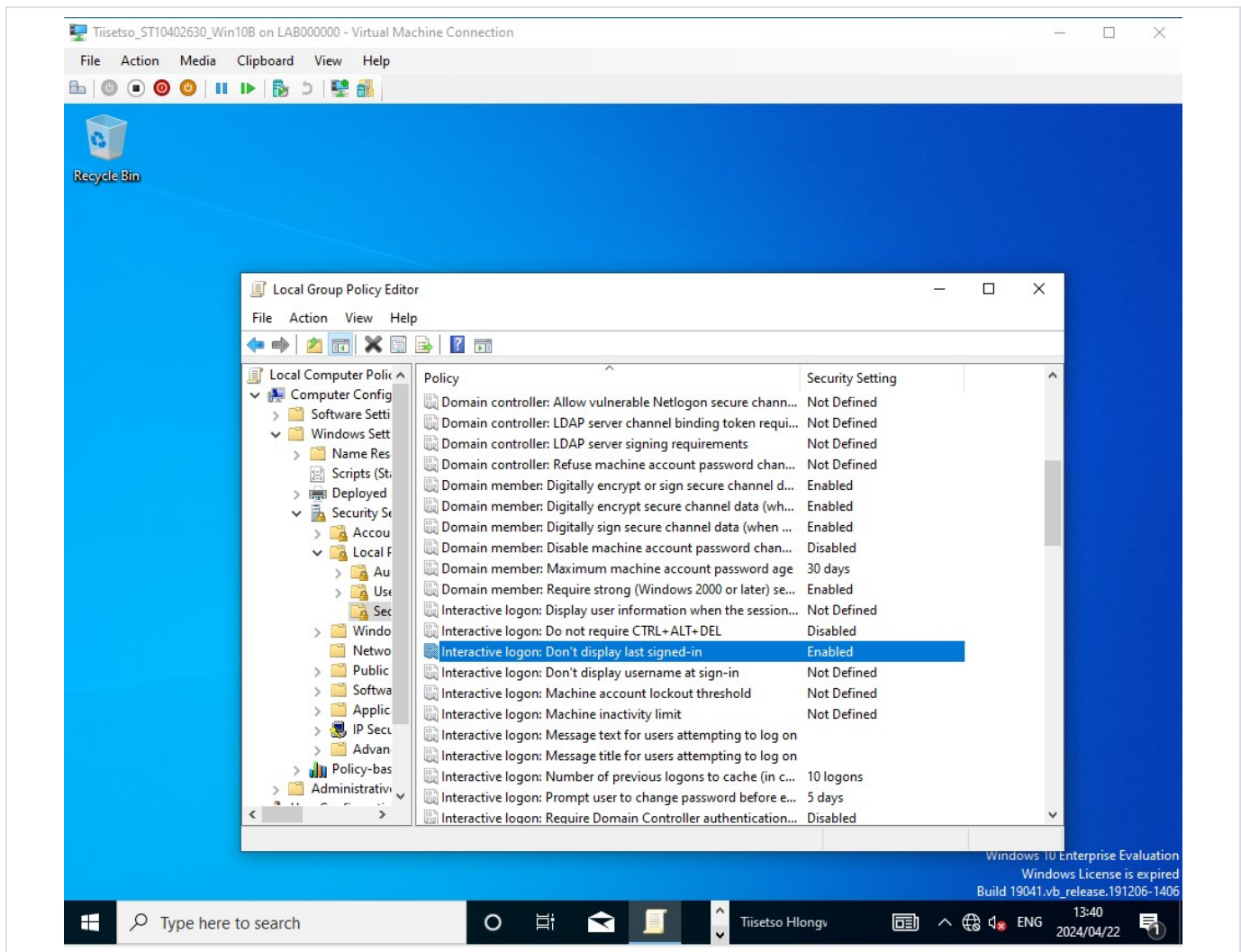
Figure 7.4: Logon Message Text Configured

Figure 7.5: Logon Message Title Configured

7.4 — Logon Screen Information Disclosure

Interactive logon: Don't display last signed-in was enabled, suppressing the previously logged-on username at the lock/logon screen. By default, Windows displays the last user who signed in — convenient, but also a small information leak: anyone standing at an unattended machine immediately learns a valid username without needing to guess or discover one, which is one half of the credential pair an attacker needs. Suppressing it costs the legitimate user nothing (they still type their own username) while removing that information from anyone else.

Figure 7.6: Last Signed-In Username Suppressed

7.5 — User Account Control (UAC) Hardening

Two UAC settings were hardened beyond the Windows default. **Only elevate executables that are signed and validated** was enabled, blocking unsigned or improperly signed executables from silently requesting administrative elevation — directly reducing the attack surface for privilege-escalation malware, which frequently relies on unsigned binaries. Separately, the elevation prompt behaviour was set to **prompt for credentials on the secure desktop**, meaning the credential prompt itself runs in an isolated desktop session that ordinary applications cannot draw over or intercept input from — a defence specifically against "UAC spoofing" attacks, where malicious software displays a fake credential dialog to harvest a password.

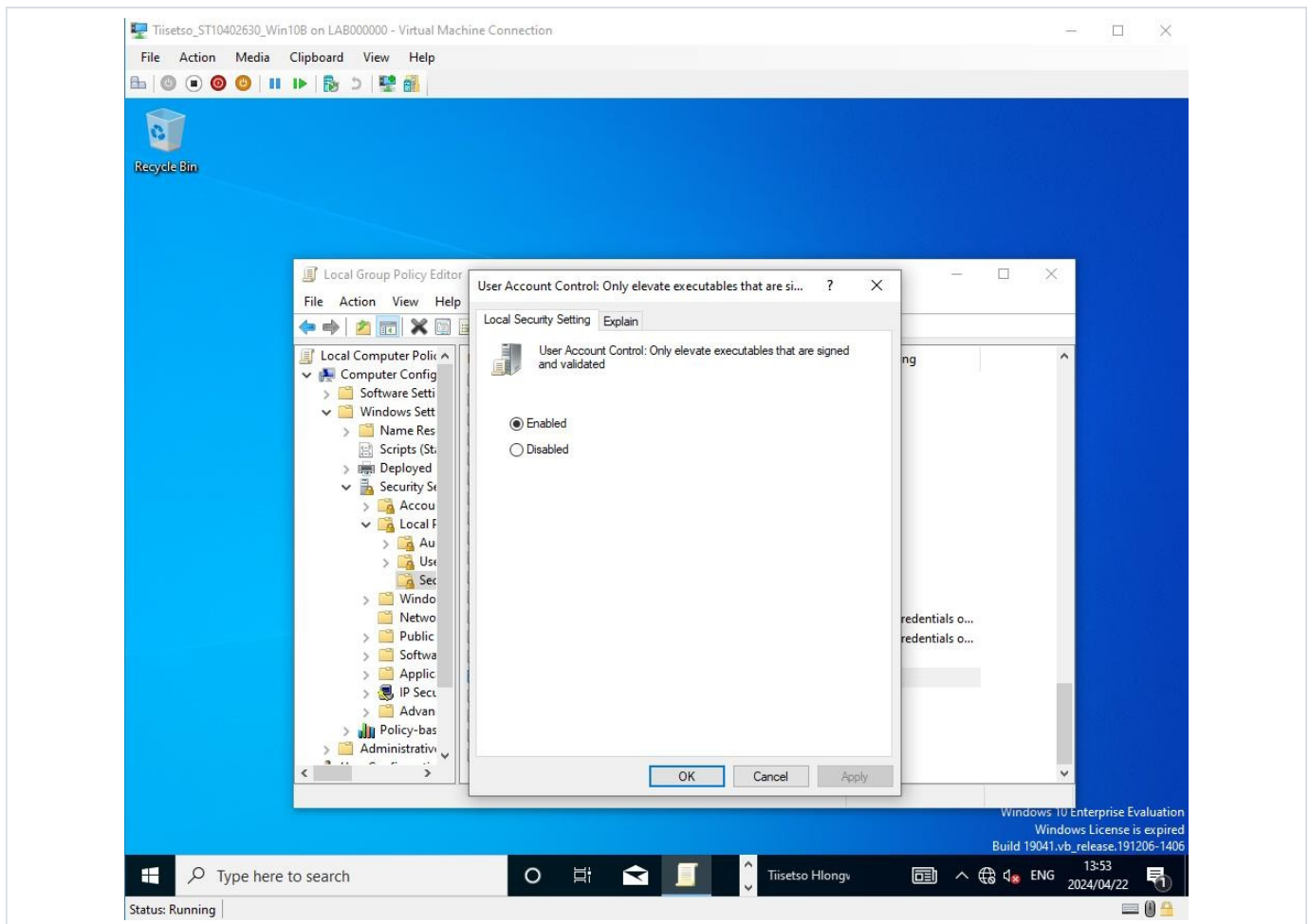
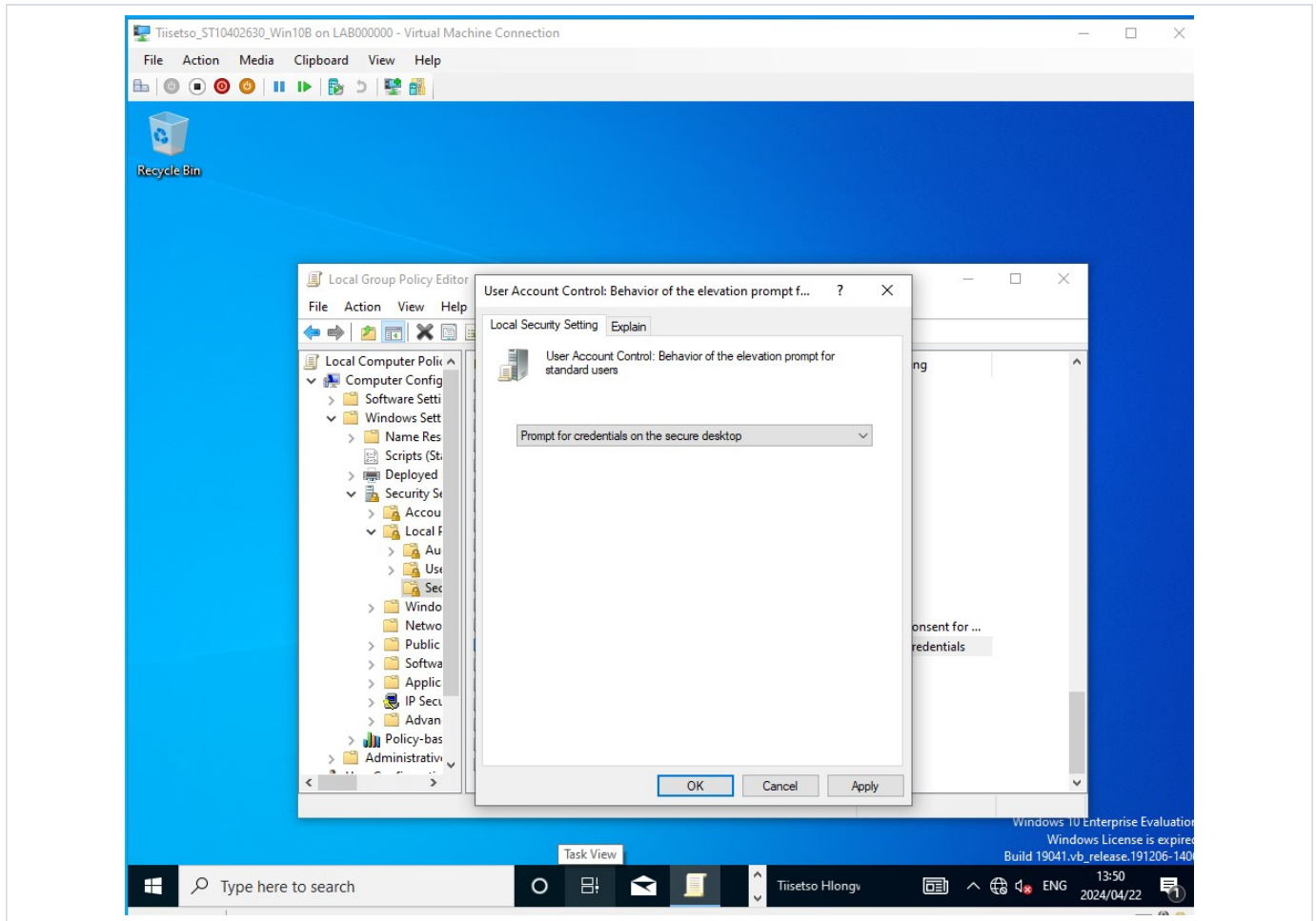
Figure 7.7: Only Elevate Signed and Validated Executables

Figure 7.8: Elevation Prompt Set to the Secure Desktop

7.6 — Policy Verification

Two final checks confirmed the security baseline was not just configured but active. First, **Windows Update** was checked and confirmed up to date, ruling out unpatched vulnerabilities as a gap in the otherwise-hardened configuration — a policy baseline is only as strong as the underlying patch level it's enforced on. Second, the machine was **restarted**, and the configured Acceptable Computer Usage Policy banner appeared exactly as configured before the logon screen — direct, functional proof that the Group Policy settings applied in this section survived a reboot and are actively enforced, rather than merely present in the policy editor.

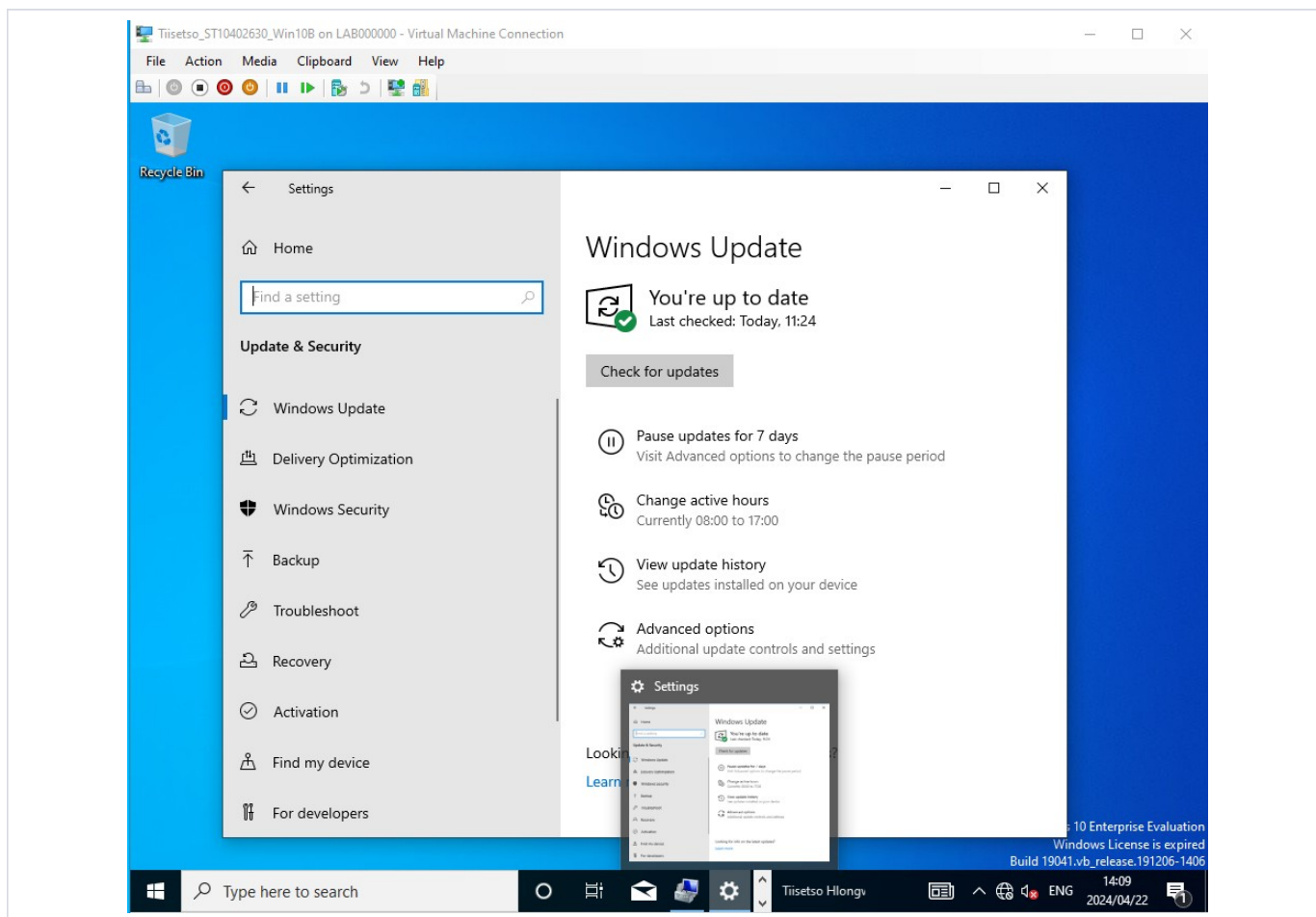
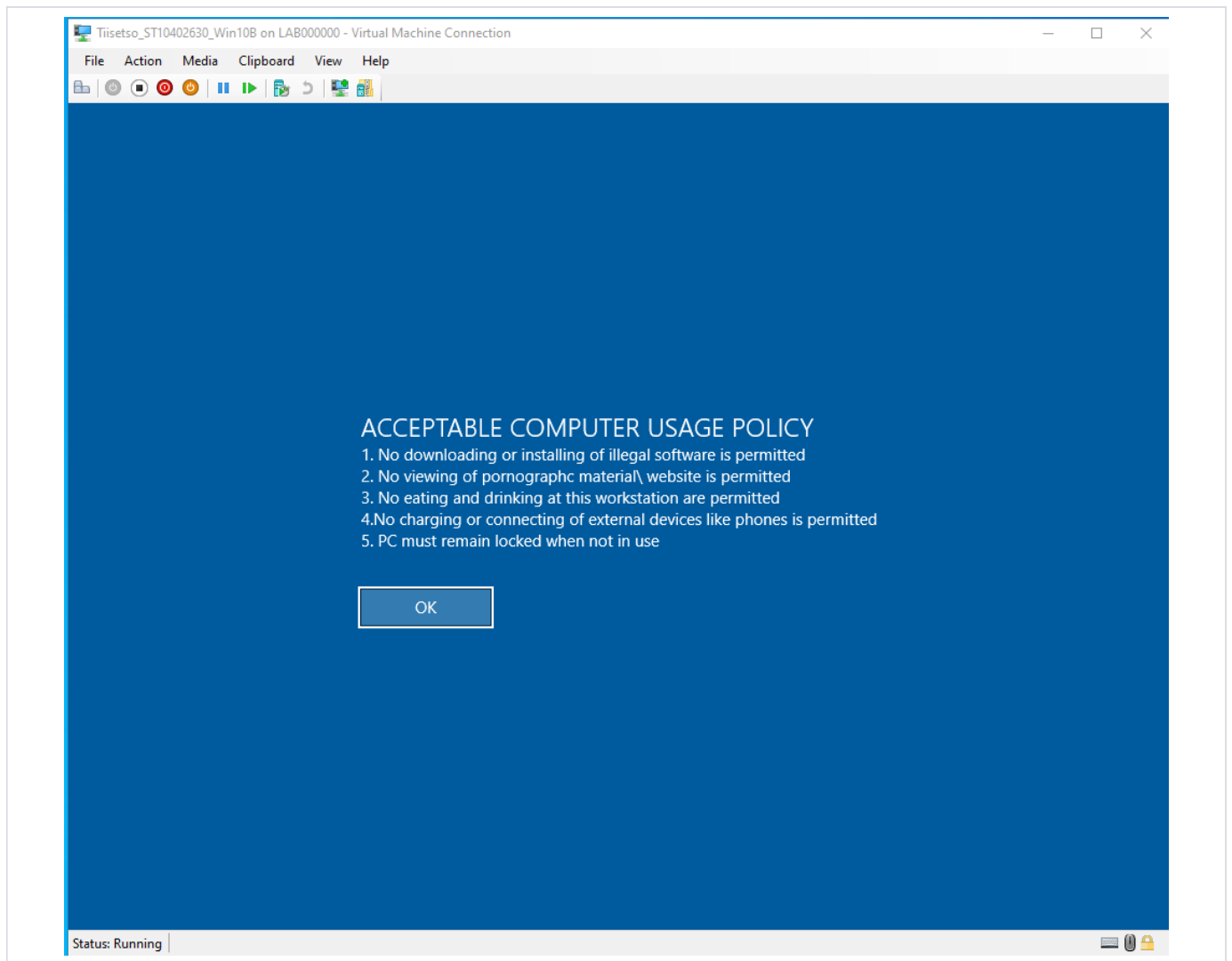
Figure 7.9: Windows Update — System Confirmed Up to Date

Figure 7.10: Logon Banner Displayed After Restart

08

SKILLS & COMPETENCIES DEMONSTRATED

Virtualization & Deployment

- › Hyper-V Generation 2 VM provisioning
- › ISO-based Windows 10 installation
- › Regional localisation configuration
- › Device identification & naming

System Utilities & Storage

- › Task Scheduler automation
- › Driver signature auditing
- › Device Manager troubleshooting
- › Storage Spaces (parity & mirror)
- › Disk & VHD management

Identity & Security Policy

- › Local user & group administration
- › Group-based privilege assignment
- › Password & lockout policy
- › UAC & logon hardening
- › Policy verification & compliance

Competency Assessment

Beginner

Expert

Foundation Level — Client Systems Administration. Demonstrated ability to provision and install a Windows 10 client operating system from a virtualised base image, operate native diagnostic and maintenance utilities correctly, configure software-defined storage resiliency with an understanding of the trade-offs between resiliency models, administer local identity through a group-based privilege model, and enforce a documented, verified local security baseline through policy rather than ad-hoc configuration.

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CONCLUSION

This portfolio documents the deployment and administration of two Windows 10 client operating systems, spanning five interconnected disciplines: virtual machine provisioning and installation, native system utilities and troubleshooting, storage resiliency configuration, local identity and privilege administration, and security policy enforcement.

The work demonstrates a consistent administrative discipline carried through every layer — infrastructure sized deliberately above the platform's minimum requirements rather than left at a bare floor; storage resiliency chosen to match the expected read/write pattern of the data it protects, and demonstrated with two different, deliberately contrasted resiliency models; identity structured around groups rather than individual accounts, so that privilege scales with membership changes instead of requiring per-user maintenance; and a security baseline — password policy, account lockout, UAC, and logon hardening — enforced through policy and confirmed working after a live restart, rather than assumed to be correct from configuration alone.

Taken together, this portfolio evidences foundation-level competency in Windows 10 client deployment and systems administration, with a demonstrated ability to take a set of platform requirements and translate them into a working, verified, and defensible technical configuration — the same underlying discipline that scales up to the domain-level, multi-site administration demonstrated in this candidate's subsequent enterprise infrastructure portfolio.

Tiisetso Tlotlo Hlongwane · CLIENT SYSTEMS ADMINISTRATION · 2024